

Discrepancy geometry in approximate Bayesian inference: transport and risk perspectives

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Abstract. Approximate Bayesian Computation (ABC) replaces the evaluation of an intractable likelihood with comparisons between observed and simulated data. We develop a unified measure-theoretic framework for discrepancy-based Bayesian inference in which such comparisons are encoded through nonnegative compatibility weights. The proposed formulation provides a mathematical setting for studying both vanishing compatibility thresholds and large-sample asymptotic regimes, leading to convergence results and an asymptotic characterization of compatibility, including situations in which posterior concentration may fail. It also admits a natural variational interpretation through an entropy-regularized minimization principle and, when the discrepancy is induced by a Wasserstein distance, an intrinsic transport representation on spaces of probability measures, giving rise to risk-theoretic functionals. These results provide a unified probabilistic, asymptotic, variational, and geometric perspective on discrepancy-based Bayesian inference.

Key words. Approximate Bayesian Computation, compatibility-based inference, generalized Bayesian inference, optimal transport, Wasserstein distance.

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Introduction

Bayesian inference traditionally relies on a tractable likelihood function relating the observed data to the underlying statistical model. In many applications, however, the data-generating mechanism can be simulated while the likelihood remains unavailable or computationally intractable. Approximate Bayesian Computation (ABC) replaces likelihood evaluation with comparisons between observed and simulated data through discrepancy measures; see, e.g., [3, 24, 33], as well as [12, 13, 7, 20]. The choice of discrepancy therefore plays a central role, influencing both the statistical behavior of posterior distributions and the structure of the inferential procedure. This has motivated the development of ABC

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methods based on distributional discrepancies, including Kullback–Leibler divergence, energy statistics, and kernel embeddings; see, e.g., [30, 21, 27]. Among these approaches, Wasserstein distances have attracted particular attention in statistics and machine learning; see, e.g., [4, 28, 37, 23].

This viewpoint naturally leads to a rigorous understanding of posterior distributions, their behavior as compatibility thresholds vanish, and the principles underlying discrepancy-based inference. To this end, we formulate Bayesian inference in terms of compatibility between the observed data and the statistical model. Compatibility is encoded through nonnegative weights measuring the agreement between observations and candidate parameter values, thus defining a family of posterior distributions. This setting allows posterior construction, vanishing-threshold limits, and large-sample regimes to be analyzed within a unified framework. Rather than constituting separate parts of the theory, these components form a coherent structure underlying the developments that follow. Two complementary perspectives then emerge. First, posterior distributions admit a natural variational interpretation linking compatibility with Gibbs variational principles. Second, the large-sample regime reveals an intrinsic transport geometry in which compatibility is expressed through Wasserstein distances. Taken together, these perspectives establish connections between Bayesian inference, optimal transport, and risk-based formulations.

The approach developed in this paper is closely related to several active directions in contemporary Bayesian inference while remaining conceptually distinct. It builds on the literature on discrepancy-based and transport-based Approximate Bayesian Computation (see, e.g., [4]) and is naturally connected with existing analyses of the asymptotic behavior of ABC posterior distributions, including under model misspecification; see, e.g., [16, 17]. It is likewise related to generalized Bayesian updating, where posterior distributions are constructed through loss or risk functionals rather than exact likelihood specifications; see, e.g., [5]. Unlike these approaches, the emphasis here is not on proposing an alternative updating rule, but on identifying the mathematical structures induced by compatibility-based inference.

The remainder of the paper unfolds these ideas progressively. Section 1 establishes the mathematical foundations of Approximate Bayesian Computation and derives the corresponding posterior construction. Section 2 introduces a general compatibility-weight formulation and characterizes the limiting posterior. Section 3 develops the asymptotic compatibility theory underlying the large-sample analysis. Section 4 explores the resulting variational and transport structures together with their risk-theoretic interpretation. Finally, Section 5 summarizes the main findings and discusses directions for future research.

1 Approximate Bayesian Computation

The aim of this section is to present a rigorous formulation of the ABC framework adopted throughout the paper. We introduce the observation and parameter spaces, the associated probabilistic model, and the discrepancy-based mechanism underlying approximate posterior inference.

1.1 A measure-theoretic framework for ABC

Fix a probability space $(\Omega, \mathcal{A}, \mathbf{P})$. Let

$$\mathcal{Y} \subseteq \mathbb{R}^{m_Y} \equiv \mathbb{R}^{m_Y \times 1}$$

be the observation space endowed with the topological structure induced by a metric d_Y , where

$$m_Y \in \mathbb{N}^* := \mathbb{N} \setminus \{0\}$$

gives the dimension of a generic data point $\mathbf{y} \in \mathcal{Y}$. We also specify the sample size $n \in \mathbb{N}^*$.

Thus, to every elementary outcome $\omega \in \Omega$ there corresponds a finite sequence of observed quantities: for every index $i = 1, \dots, n$, the i -th observation corresponding to ω is denoted by

$$\mathbf{y}^i \equiv \mathbf{y}^i(\omega) \in \mathcal{Y}.$$

These observations are collectively represented by the sample vector at ω , defined as

$$\mathbf{y}^{1:n} \equiv \mathbf{y}^{1:n}(\omega) := [\mathbf{y}^1(\omega), \dots, \mathbf{y}^n(\omega)] \in \mathcal{Y}^n \subseteq \mathbb{R}^{m_Y \times n}.$$

In what follows, the dependence upon ω will be tacitly omitted whenever no ambiguity arises, so that the notation $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ is understood as denoting a fixed realization of the observed sample.

This construction implicitly gives rise to mappings from Ω to $\mathcal{Y}$ and to $\mathcal{Y}^n$, respectively; however, no properties of these mappings, such as measurability or further regularity, will be required.

We next introduce the parameter space

$$\Theta \subseteq \mathbb{R}^{m_\Theta},$$

governing the data-generating mechanism and equipped with the topology induced by a metric d_Θ , where

$$m_\Theta \in \mathbb{N}^*$$

is the dimension of the parameter vectors $\vartheta \in \Theta$.

Notation. For any topological space X , we denote by $\mathcal{B}(X)$ its Borel σ -algebra and by $\mathcal{P}(X)$ the set of probability measures defined on $\mathcal{B}(X)$. Given another topological space Y , we write $\mathcal{B}(X, Y)$ for the class of Borel measurable mappings from X to Y .

We further consider a prior distribution

$$\pi \in \mathcal{P}(\Theta)$$

on the parameter space Θ , together with the associated family of parametric models

$$\{\mu_\vartheta^n\}_{\vartheta \in \Theta} \subseteq \mathcal{P}(\mathcal{Y}^n),$$

where, for each $\vartheta \in \Theta$, the measure μ_ϑ^n represents the distribution of the sample vector under ϑ .

The first structural assumption underlying Approximate Bayesian Computation can be stated and will henceforth be maintained throughout the paper.

(A-1) — *Simulatability*. The statistical model $\{\mu_\vartheta^n\}_{\vartheta \in \Theta}$ is generative in the sense that, for π -almost every $\vartheta \in \Theta$, one can simulate arbitrarily many independent pseudo-samples

$$\mathbf{y}_\vartheta^{1:n} \equiv \mathbf{y}_\vartheta^{1:n}(\omega) := [\mathbf{y}_\vartheta^1(\omega), \dots, \mathbf{y}_\vartheta^n(\omega)] \in \mathcal{Y}^n,$$

namely random vectors distributed according to μ_ϑ^n , that is, $\mathbf{y}_\vartheta^{1:n} \sim \mu_\vartheta^n$.

In view of (A-1), for π -almost every $\vartheta \in \Theta$, let

$$\mathcal{Y}_\vartheta^n \in \mathcal{B}(\mathcal{Y}^n) \tag{1}$$

denote a full-measure set for the pseudo-samples generated under parameter ϑ , in the sense that

$$\mu_\vartheta^n[\mathcal{Y}_\vartheta^n] = 1.$$

Heuristically, the set $\mathcal{Y}_\vartheta^n$ may be interpreted as the collection of all pseudo-samples distributed according to μ_ϑ^n , up to measurable subsets of $\mathcal{Y}^n$ of μ_ϑ^n -measure zero, since probabilistic statements formally concern random samples rather than individual deterministic realizations.

We emphasize that the observed sample vector $\mathbf{y}^{1:n}$ constitutes the fixed empirical realization arising from the underlying statistical experiment and is regarded as given, whereas any pseudo-sample $\mathbf{y}_\vartheta^{1:n}$ corresponds to simulated data generated from the model under a prescribed parameter value $\vartheta \in \Theta$. Pseudo-samples thus form variable elements of the observation space $\mathcal{Y}^n$, whose probability distribution is entirely determined by the underlying statistical model.

This distinction lies at the heart of ABC: in the absence of direct likelihood evaluation, inference compares observed and simulated data through a discrepancy measure

$$\mathcal{D}: \mathcal{Y}^n \times \mathcal{Y}^n \longrightarrow \mathbb{R}_+ := [0, \infty[,$$

defined as a continuous pseudo-metric on $\mathcal{Y}^n$, satisfying all the standard axioms of a metric except that distinct elements may have zero distance; see, e.g., [30, 4, 27].

To this end, for any observed sample $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ and radius $r \in \mathbb{R}_+$, we introduce the sample-dependent closed subset $D_r(\mathbf{y}^{1:n})$ of the observation space $\mathcal{Y}^n$ containing all elements whose deviation from $\mathbf{y}^{1:n}$ does not exceed r under the discrepancy measure $\mathcal{D}$; namely,

$$D_r(\mathbf{y}^{1:n}) := \mathcal{D}(\mathbf{y}^{1:n}, \cdot)^{-1}([0, r]) = \{\tilde{\mathbf{y}}^{1:n} \in \mathcal{Y}^n \mid \mathcal{D}(\mathbf{y}^{1:n}, \tilde{\mathbf{y}}^{1:n}) \leq r\} \in \mathcal{B}(\mathcal{Y}^n).$$

The family $\{D_r(\mathbf{y}^{1:n})\}_{r \geq 0}$ is increasing in r and, as $r \downarrow 0$,

$$D_r(\mathbf{y}^{1:n}) \downarrow D_0(\mathbf{y}^{1:n}) \equiv \mathcal{D}(\mathbf{y}^{1:n}, \cdot)^{-1}(\{0\}).$$

We are in a position to formulate the second fundamental assumption of Approximate Bayesian Computation, which will be adopted from this point onward.

($\mathcal{A}$ -2) — *Nondegeneracy*. There exists $\varepsilon^\circ \in]0, \infty[$ such that, for every $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ and every $\varepsilon \in]0, \varepsilon^\circ[$, hereafter referred to as an ABC threshold, the mapping from Θ to $[0, 1]$ given by

$$\vartheta \mapsto \mu_\vartheta^n[D_\varepsilon(\mathbf{y}^{1:n})] \quad (2)$$

belongs to $\mathcal{B}(\Theta, [0, 1])$ and is not π -almost surely equal to zero.

For each fixed $\mathbf{y}^{1:n} \in \mathcal{Y}^n$, ABC threshold ε , and π -almost every $\vartheta \in \Theta$, the quantity $\mu_\vartheta^n[D_\varepsilon(\mathbf{y}^{1:n})]$ represents the probability that pseudo-samples $\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n$ generated under the parameter ϑ fall within the ε -neighborhood $D_\varepsilon(\mathbf{y}^{1:n})$ of the observed sample; that is, the probability that simulated data are sufficiently close to the empirical realization in the sense induced by the discrepancy measure $\mathcal{D}$.

When regarded as a function of the parameter ϑ , as in (2), it defines a measure of compatibility and can be interpreted as a surrogate likelihood, in which the classical pointwise comparison between observed and model-implied data is replaced by a tolerance-based notion of proximity; see, e.g., [24, 33]. Correspondingly, large values indicate that the model with ϑ assigns high probability to realizations close to the observed sample, while small values reflect poor compatibility with the reference sample $\mathbf{y}^{1:n}$.

Remark 1.1. As an immediate consequence of (($\mathcal{A}$ -1) and) ($\mathcal{A}$ -2), for every $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ and every ABC threshold ε , the function (2) is π -integrable (in the Lebesgue sense), and

$$\int_{\Theta} \mu_\vartheta^n[D_\varepsilon(\mathbf{y}^{1:n})] \pi(d\vartheta) \in]0, 1].$$

1.2 ABC posterior distribution

We now turn to the probabilistic formulation of Approximate Bayesian Computation. We define a probability measure on the measurable space $(\Theta, \mathcal{B}(\Theta))$ —that is, an element of $\mathcal{P}(\Theta)$ —which serves as an approximate Bayesian posterior distribution when the likelihood function is not available in closed form, but simulation from the model remains feasible (according to ($\mathcal{A}$ -1)).

The core idea is to replace exact matching between observed and simulated data by quantifying their discrepancy through the measure $\mathcal{D}$. For π -almost every $\vartheta \in \Theta$, pseudo-samples $\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n$ are generated from the model and compared to the observed sample $\mathbf{y}^{1:n}$ using $\mathcal{D}(\mathbf{y}^{1:n}, \mathbf{y}_\vartheta^{1:n})$.

Given an ABC threshold ε , this leads to a reweighting of the prior distribution π , where each parameter value ϑ is weighted according to the probability that the corresponding pseudo-samples fall within the neighborhood $D_\varepsilon(\mathbf{y}^{1:n})$. The resulting probability measure on Θ can thus be interpreted as an approximate Bayesian update and is formalized below as the ABC posterior distribution, in which the role traditionally played by the likelihood is taken by the compatibility map (2); see, e.g., [24, 33].

The following scheme describes the classical ABC rejection algorithm; see [3, 24].

Algorithm 1 (ABC rejection algorithm). *Let $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ be an observed sample.*

Step 1 (Tolerance specification). *Choose an ABC threshold $\varepsilon \in]0, \varepsilon^\circ[$.*

Step 2 (Sampling and simulation). *Draw a parameter $\vartheta \in \Theta$ from the prior distribution π and generate a pseudo-sample $\mathbf{y}_{\vartheta}^{1:n} \in \mathcal{Y}_{\vartheta}^n$.*

Step 3 (Acceptance rule). *Retain ϑ if and only if $\mathbf{y}_{\vartheta}^{1:n} \in D_{\varepsilon}(\mathbf{y}^{1:n})$.*

By iterating steps 2–3 of Algorithm 1, one successively generates accepted parameter values distributed according to the approximate Bayesian mechanism described above.

More precisely, the parameter values retained at Step 3 follow the ABC posterior distribution

$$\pi_{\varepsilon}^n \equiv \pi_{\varepsilon}^{\mathbf{y}^{1:n}} \ll \pi, \quad \varepsilon \in]0, \varepsilon^{\circ}[,$$

whose Radon–Nikodym derivative with respect to π is, by construction, proportional to the mapping (2) (see also Remark 1.1): that is, for every $B \in \mathcal{B}(\Theta)$,

$$\pi_{\varepsilon}^n[B] = \frac{\int_B \mu_{\vartheta}^n[D_{\varepsilon}(\mathbf{y}^{1:n})] \pi(d\vartheta)}{\int_{\Theta} \mu_{\vartheta'}^n[D_{\varepsilon}(\mathbf{y}^{1:n})] \pi(d\vartheta')}, \quad (3)$$

Remark 1.2. In light of the foregoing, for every $r \in \mathbb{R}_+$ and π -almost every $\vartheta \in \Theta$, the set

$$D_r(\mathbf{y}^{1:n}) \cap \mathcal{Y}_{\vartheta}^n \subseteq \mathcal{Y}^n$$

may be interpreted as the acceptance region associated with the tolerance level r , consisting of all pseudo-samples sufficiently close to the observed data. The corresponding acceptance probability is given by $\mu_{\vartheta}^n[D_r(\mathbf{y}^{1:n})]$, namely the probability that a sample generated under the parameter value ϑ is accepted, thus quantifying the degree of compatibility between the model at ϑ and the observed sample.

2 On the vanishing-threshold limit

In this section, we develop a structural setting to analyze the asymptotic behavior of approximate Bayesian procedures as the tolerance parameter ε vanishes, while keeping the sample size n fixed, consistently with the previous section. Specifically, we formulate conditions under which the approximate posterior distribution (3) admits a limit as $\varepsilon \downarrow 0$, and characterize that limit.

The complementary regime of increasing sample size n is addressed in Section 3, where the analysis requires a richer probabilistic and geometric framework, as the posterior is governed not only by the tolerance, but also by the discrepancy geometry arising on spaces of probabilistic representations.

2.1 A general convergence principle

Consider a general family of nonnegative measurable functions

$$w_{\varepsilon} \in \mathcal{B}(\Theta, \mathbb{R}_+)$$

indexed by a parameter $\varepsilon \in]0, \infty[$, intended to measure the compatibility between a parameter $\vartheta \in \Theta$ and the observed sample $\mathbf{y}^{1:n}$, in the same spirit as (2); throughout, we refer to them as compatibility weights. We assume that, for every sufficiently small ε , w_{ε} is π -integrable and not π -almost surely equal to zero. Accordingly, these functions naturally induce the family $\{\pi_{\varepsilon}\}_{\varepsilon}$ in $\mathcal{P}(\Theta)$ defined by

$$\pi_{\varepsilon}[B] := \frac{\int_B w_{\varepsilon}(\vartheta) \pi(d\vartheta)}{\int_{\Theta} w_{\varepsilon}(\vartheta') \pi(d\vartheta')}, \quad B \in \mathcal{B}(\Theta). \quad (4)$$

This formulation encompasses the classical ABC posterior (3) as a special case; see, e.g., [3, 24, 33].

We next establish a fundamental convergence principle governed by the asymptotic behavior of the compatibility weights w_{ε} in the vanishing-threshold regime.

Proposition 2.1. Assume that there exist a family $\{\kappa_\varepsilon\}_\varepsilon$ of strictly positive constants and a function $\bar{w} \in \mathcal{B}(\Theta, \mathbb{R}_+)$, π -integrable and not π -almost surely equal to zero, such that

$$\lim_{\varepsilon \downarrow 0} \int_{\Theta} \left| \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) - \bar{w}(\vartheta) \right| \pi(d\vartheta) = 0. \quad (5)$$

Then, as $\varepsilon \downarrow 0$, the family of compatibility-weight posterior distributions $\{\pi_\varepsilon\}_\varepsilon$ introduced in (4) converges in total variation to the probability measure $\bar{\pi} \in \mathcal{P}(\Theta)$ defined by

$$\bar{\pi}[B] := \frac{\int_B \bar{w}(\vartheta) \pi(d\vartheta)}{\int_{\Theta} \bar{w}(\vartheta') \pi(d\vartheta')}, \quad B \in \mathcal{B}(\Theta), \quad (6)$$

that is,

$$\lim_{\varepsilon \downarrow 0} \sup_{B \in \mathcal{B}(\Theta)} |\pi_\varepsilon[B] - \bar{\pi}[B]| = 0. \quad (7)$$

Proof. For every $B \in \mathcal{B}(\Theta)$, by the definitions above and the standard triangle inequality for integrals,

$$\begin{aligned} & |\pi_\varepsilon[B] - \bar{\pi}[B]| \\ &= \left| \frac{\int_B \left[\frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) - \bar{w}(\vartheta) \right] \pi(d\vartheta)}{\int_{\Theta} \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta') \pi(d\vartheta')} + \left[\frac{1}{\int_{\Theta} \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta') \pi(d\vartheta')} - \frac{1}{\int_{\Theta} \bar{w}(\vartheta') \pi(d\vartheta')} \right] \int_B \bar{w}(\vartheta) \pi(d\vartheta) \right| \\ &\leq \frac{\int_{\Theta} \left| \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) - \bar{w}(\vartheta) \right| \pi(d\vartheta)}{\int_{\Theta} \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta') \pi(d\vartheta')} + \left| \frac{1}{\int_{\Theta} \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta') \pi(d\vartheta')} - \frac{1}{\int_{\Theta} \bar{w}(\vartheta') \pi(d\vartheta')} \right| \int_{\Theta} \bar{w}(\vartheta) \pi(d\vartheta) \end{aligned}$$

and each term on the right-hand side converges to zero as $\varepsilon \downarrow 0$ as a direct consequence of (5). The convergence is uniform with respect to $B \in \mathcal{B}(\Theta)$, which proves (7). $\square$

Within the present framework, Proposition 2.1 highlights that the asymptotic behavior of the induced posterior measures π_ε is essentially governed by the corresponding rescaled compatibility weights $w_\varepsilon/\kappa_\varepsilon$. In particular, sufficiently strong convergence in integral form of such weights— $L^1(\pi)$ -convergence—yields convergence in total variation toward a well-defined limiting posterior distribution $\bar{\pi}$.

This compatibility-weight formulation also underlies the variational and transport formulations of approximate Bayesian inference discussed in detail in Section 4.

2.2 Classical ABC as a particular case

The general approach developed above applies to a broad class of approximate Bayesian constructions. We specialize it to the classical ABC setting, where the compatibility weights are induced by acceptance probabilities over shrinking neighborhoods of the observed sample; see, e.g., [3, 24, 33].

For this purpose, let $\mathbf{y}^{1:n} \in \mathcal{Y}^n$ be the observed sample and assume that, for π -almost every $\vartheta \in \Theta$, the measure μ_ϑ^n is absolutely continuous with respect to the $nm_{\mathcal{Y}}$ -dimensional Lebesgue measure

$$\mathbf{Vol}_{nm_{\mathcal{Y}}} := \lambda^{nm_{\mathcal{Y}}}$$

restricted to $\mathcal{B}(\mathcal{Y}^n)$, with Radon–Nikodym derivative

$$f_\vartheta^n := \frac{d\mu_\vartheta^n}{d\mathbf{Vol}_{nm_{\mathcal{Y}}}}$$

admitting a version, still denoted by f_ϑ^n , which is continuous at $\mathbf{y}^{1:n}$. Moreover, define

$$\bar{w}(\vartheta) := f_\vartheta^n(\mathbf{y}^{1:n}) \quad (8)$$

and suppose that it belongs to $\mathcal{B}(\Theta, \mathbb{R}_+)$ and is not π -almost surely equal to zero, so that it may be interpreted as the likelihood function associated with $\mathbf{y}^{1:n}$. Accordingly, for every ABC threshold ε , let

$$\kappa_\varepsilon := \mathbf{Vol}_{nmy} [D_\varepsilon(\mathbf{y}^{1:n})]$$

and, for π -almost every $\vartheta \in \Theta$,

$$w_\varepsilon(\vartheta) := \mu_\vartheta^n [D_\varepsilon(\mathbf{y}^{1:n})] = \int_{D_\varepsilon(\mathbf{y}^{1:n})} f_\vartheta^n(\mathbf{z}^{1:n}) d\mathbf{z}^{1:n}.$$

Under the additional assumptions that $\mathcal{D}$ is a genuine metric compatible with the topology of $\mathcal{Y}^n$ and $0 < \kappa_\varepsilon < \infty$ for every sufficiently small ε , it follows that, for π -almost every $\vartheta \in \Theta$,

$$\lim_{\varepsilon \downarrow 0} \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) = \bar{w}(\vartheta)$$

by continuity of f_ϑ^n at $\mathbf{y}^{1:n}$ and the standard local averaging property over shrinking metric balls.

Therefore, if there exists a function $g \in \mathcal{B}(\Theta, \mathbb{R}_+)$ which is π -integrable and such that, for every sufficiently small ε and for π -almost every $\vartheta \in \Theta$,

$$\frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) \leq g(\vartheta),$$

then (5) follows from the Lebesgue dominated convergence theorem. Consequently, Proposition 2.1 yields convergence in total variation of the ABC posterior $\pi_\varepsilon \equiv \pi_\varepsilon^n$ toward the posterior distribution (6) associated with the likelihood (8), explicitly given by

$$\bar{\pi}[B] = \frac{\int_B f_\vartheta^n(\mathbf{y}^{1:n}) \pi(d\vartheta)}{\int_\Theta f_{\vartheta'}^n(\mathbf{y}^{1:n}) \pi(d\vartheta')}, \quad B \in \mathcal{B}(\Theta).$$

From a measure-theoretic perspective, the limiting measure above coincides with the usual likelihood-based posterior distribution associated with the observation $\mathbf{y}^{1:n}$, thus recovering the classical Bayesian update. Its density with respect to π is induced by the function (8); see, e.g., [29].

Altogether, the previous analysis shows that classical ABC convergence results emerge as a particular manifestation of Proposition 2.1. More precisely, the fundamental mechanism underlying posterior convergence is not tied to the specific structure of ABC neighborhoods themselves, but rather to an $L^1(\pi)$ -type control of suitably normalized compatibility weights.

Within this framework, uniform local bounds on the likelihood are not intrinsically required and may be replaced by an integrated domination condition with respect to the prior distribution π . The convergence result remains applicable even in settings where the likelihood exhibits substantial local variability, provided that its averaged behavior retains sufficient π -integrability.

This viewpoint suggests that several classical assumptions commonly imposed in the ABC literature (see, e.g., [24, 33]) should be understood primarily as sufficient conditions ensuring the necessary integral control, rather than as intrinsic structural requirements for posterior convergence.

3 The large-sample regime

The purpose of this section is to show how discrepancy-based acceptance mechanisms give rise in the large-sample limit to a non-trivial asymptotic compatibility structure over the parameter space, instead of treating posterior consistency in the classical parametric sense as the primary objective.

Under this regime and for a fixed ABC tolerance, ABC posterior distributions need not asymptotically concentrate on the minimizers of the limiting discrepancy function. This is particularly relevant in realistic inference settings, where posterior concentration arguments often rely on identifiability, specification, or sufficiency assumptions that are frequently unavailable in practice; see, e.g., [16, 17].

From this perspective, ABC procedures should not be interpreted merely as imperfect approximations of exact Bayesian inference, but as discrepancy-induced asymptotic compatibility structures governed by the compatibility geometry, not by parametric identification.

3.1 Probabilistic representations and asymptotic discrepancies

The construction below separates the sampling mechanism of the ABC procedure from the asymptotic discrepancy geometry governing acceptance. Rather than working with finite-dimensional sample vectors, we represent observed and simulated samples by probability measures. This is particularly appropriate in the large-sample regime, since the ambient sample spaces vary with n , while these probability measures evolve in a common space. Acceptance is then expressed in terms of the discrepancy between them.

Let

$$\mathcal{T} : \mathcal{P}(\mathcal{Y}) \times \mathcal{P}(\mathcal{Y}) \longrightarrow \mathbb{R}_+ \quad (9)$$

denote a generic pseudo-metric defined on $\mathcal{P}(\mathcal{Y})$. At the present level of generality, the discrepancy map $\mathcal{T}$ naturally encompasses transport structures, including optimal-transport geometries, whose role in discrepancy-based inference will be further explored in Section 4; see, e.g., [4].

For every sample size $n \in \mathbb{N}^*$ and for $\mathbf{P}$ -almost every $\omega, \omega' \in \Omega$, we associate to the observed sample $\mathbf{y}^{1:n} \equiv \mathbf{y}^{1:n}(\omega) \in \mathcal{Y}^n$ a probability measure

$$\nu_n \equiv \nu_n(\mathbf{y}^{1:n}) \in \mathcal{P}(\mathcal{Y})$$

and similarly, for π -almost every $\vartheta \in \Theta$, to each pseudo-sample $\mathbf{y}_\vartheta^{1:n} \equiv \mathbf{y}_\vartheta^{1:n}(\omega') \in \mathcal{Y}_\vartheta^n$, generated under parameter ϑ , a probability measure

$$\nu_{\vartheta,n} \equiv \nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}) \in \mathcal{P}(\mathcal{Y}).$$

A natural choice is given by empirical measures, although the subsequent analysis does not rely on this specific representation; see, e.g., [4, 28]. What is essential is that observed and simulated samples admit a common probabilistic representation through which the asymptotic behavior of the ABC acceptance mechanism can be characterized in terms of the geometry induced by $\mathcal{T}$.

Remark 3.1. Since $\mathcal{T}$ is only assumed to be a pseudo-metric, the topology induced on $\mathcal{P}(\mathcal{Y})$ need not be Hausdorff (equivalently, T_2). In particular, distinct probability measures on $\mathcal{Y}$ may fail to be topologically separated whenever their discrepancy with respect to $\mathcal{T}$ vanishes.

We next assume that, for every $n \in \mathbb{N}^*$, every $\mathbf{y}^{1:n} \in \mathcal{Y}^n$, and π -almost every $\vartheta \in \Theta$, the mapping

$$\mathbf{y}_\vartheta^{1:n} \longmapsto \Delta_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n}) := \mathcal{T}(\nu_n, \nu_{\vartheta,n}) \equiv \mathcal{T}(\nu_n(\mathbf{y}^{1:n}), \nu_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n})) \quad (10)$$

belongs to $\mathcal{B}(\mathcal{Y}_\vartheta^n, \mathbb{R}_+)$ (see (1)).

The large-sample behavior of the ABC compatibility selection mechanism is ultimately driven not by the finite-dimensional samples, but by the limiting geometry induced by their probabilistic representations. Motivated by this observation, we now pass to the asymptotic counterpart and assume that the observed probability measures converge, with respect to $\mathcal{T}$, toward a limiting law.

More precisely, we assume the existence of a probability measure

$$\mu^\star \in \mathcal{P}(\mathcal{Y}) \quad (11)$$

such that, for every $n \in \mathbb{N}^*$, the mapping

$$\omega \longmapsto \mathcal{T}_n^\star := \mathcal{T}(\mu^\star, \nu_n) \equiv \mathcal{T}(\mu^\star, \nu_n(\mathbf{y}^{1:n}(\omega))) \quad (12)$$

is $\mathcal{A}$ -measurable and converges to zero in $\mathbf{P}$ -probability as $n \rightarrow \infty$, namely

$$\mathcal{T}_n^\star \xrightarrow[n \rightarrow \infty]{\mathbf{P}} 0. \quad (13)$$

This condition is fulfilled, for instance, whenever the convergence holds $\mathbf{P}$ -almost surely.

Similarly, for π -almost every $\vartheta \in \Theta$, we assume the existence of a probability measure

$$\nu_\vartheta \in \mathcal{P}(\mathcal{Y}) \quad (14)$$

such that the function

$$\vartheta \mapsto \mathcal{T}_\vartheta^\star := \mathcal{T}(\mu^\star, \nu_\vartheta) \quad (15)$$

belongs to $\mathcal{B}(\Theta, \mathbb{R}_+)$ and, for every $n \in \mathbb{N}^*$ and π -almost every $\vartheta \in \Theta$,

$$\mathbf{y}_\vartheta^{1:n} \mapsto \mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta)$$

belongs to $\mathcal{B}(\mathcal{Y}_\vartheta^n, \mathbb{R}_+)$. This allows us to associate with π -almost every parameter ϑ an asymptotic discrepancy level relative to the limiting observational law, namely $\mathcal{T}_\vartheta^\star$. The resulting functional (15) captures the compatibility geometry of the model class induced by the observed probabilistic regime and may be interpreted as an asymptotic discrepancy function over the parameter space.

Finally, we assume that the π -essential infimum level of asymptotic discrepancy coincides with the ordinary minimum and is attained by at least one parameter. More precisely, we set

$$\varepsilon^\star := \operatorname{ess\,inf}_{\vartheta \in \Theta} \mathcal{T}_\vartheta^\star = \min_{\vartheta \in \Theta} \mathcal{T}_\vartheta^\star \quad (16)$$

and define the nonempty measurable set

$$\Theta^\star := \{\vartheta^\star \in \Theta \mid \mathcal{T}_{\vartheta^\star}^\star = \varepsilon^\star\} \neq \emptyset. \quad (17)$$

The quantity $\varepsilon^\star$ represents the smallest discrepancy level asymptotically achievable within the model class and serves as the fundamental compatibility threshold throughout the sequel.

Throughout the remainder of the section, we assume that this minimal compatibility level lies strictly below the maximal admissible ABC tolerance ε° ; that is,

$$\varepsilon^\star < \varepsilon^\circ. \quad (18)$$

Accordingly, all subsequent excess thresholds ε are implicitly assumed to satisfy

$$\varepsilon \in]0, \varepsilon^\circ - \varepsilon^\star[, \quad (19)$$

which ensures the admissibility of the effective operational tolerance level

$$\varepsilon^\star + \varepsilon.$$

Two qualitatively different asymptotic regimes may arise. The case $\varepsilon^\star = 0$ corresponds to asymptotic compatibility between the observed probabilistic regime and the model class, whereas $\varepsilon^\star > 0$ reflects an intrinsic asymptotic discrepancy that may emerge under model misspecification, structural incompatibility, or limited expressive capacity of the simulated probabilistic family; see, e.g., [17].

Remark 3.2. Although technically structured in appearance, the assumptions introduced above are fulfilled in a broad range of standard Approximate Bayesian Computation settings.

This includes discrepancy constructions induced by Wasserstein distances and kernel-induced metrics such as the maximum mean discrepancy, as well as functionals built from measurable summary statistics; see, e.g., [34, 30, 4]. Related approaches based on other probability metrics, including Prokhorov or total variation discrepancies, may also be accommodated whenever the chosen probabilistic representations satisfy the required measurability and convergence properties. Under the corresponding regularity conditions, the dependence of these discrepancies on the simulated pseudo-samples is measurable and, in many practical situations, continuous with respect to the underlying sample variables.

The probability measures $\mu^\star$ and ν_ϑ may arise as empirical limits under familiar probabilistic regimes, including i.i.d. sampling, ergodic observations, and suitable weak-convergence settings, provided that $\mathcal{T}$ is compatible with the corresponding mode of convergence; see, e.g., [6, 15].

Finally, the equality between the essential infimum and the minimum in (16) is guaranteed under appropriate compactness, continuity, and full-support assumptions.

Overall, these observations indicate that the asymptotic function induced by (15) should be regarded not merely as an abstract construction, but rather as a natural geometric object encompassing many discrepancy-based methodologies commonly employed in modern ABC theory and practice.

3.2 Acceptance geometry and posterior compatibility

The asymptotic discrepancy structure generates a geometric acceptance mechanism over the parameter space. Following the acceptance criterion underlying ABC (see Algorithm 1), pseudo-samples $\mathbf{y}_{\vartheta}^{1:n} \in \mathcal{Y}_{\vartheta}^n$ generated under $\vartheta \in \Theta$ are retained whenever their probabilistic representation $\nu_{\vartheta,n}$ lies sufficiently close to the law ν_n induced by the observed sample $\mathbf{y}^{1:n} \in \mathcal{Y}^n$, relative to the minimal asymptotic discrepancy level ε^* given in (16). This level acts as the baseline for acceptance in view of (18).

This gives rise, through the measurable discrepancy map (10), to the finite-sample μ^* -compatibility acceptance region, for every admissible excess threshold ε in the sense of (19), defined by

$$A_{\vartheta,n}(\varepsilon) := \{ \mathbf{y}_{\vartheta}^{1:n} \in \mathcal{Y}_{\vartheta}^n \mid \Delta_{\vartheta,n}(\mathbf{y}_{\vartheta}^{1:n}) \leq \varepsilon^* + \varepsilon \} \in \mathcal{B}(\mathcal{Y}_{\vartheta}^n), \quad (20)$$

for every $n \in \mathbb{N}^*$, every observed sample $\mathbf{y}^{1:n} \in \mathcal{Y}^n$, and π -almost every parameter $\vartheta \in \Theta$.

Consistently with Remark 1.2 and Section 2, the quantity

$$w_{\varepsilon}^n(\vartheta) := \mu_{\vartheta}^n[A_{\vartheta,n}(\varepsilon)] \quad (21)$$

may then be viewed as the compatibility weight induced by the acceptance geometry at effective tolerance level $\varepsilon^* + \varepsilon$ for the parameter ϑ . In probabilistic terms, it represents the acceptance probability of pseudo-samples generated under ϑ relative to the observed data-generating regime.

In what follows, we implicitly assume that, for every excess threshold ε and all $n \in \mathbb{N}^*$, the mapping

$$\vartheta \mapsto w_{\varepsilon}^n(\vartheta)$$

belongs to $\mathcal{B}(\Theta, [0, 1])$ and is not π -almost surely equal to zero. This excludes degenerate situations in which the compatibility mechanism has zero prior-averaged acceptance probability.

The resulting normalization constant is given by

$$C_{\varepsilon}^n := \int_{\Theta} w_{\varepsilon}^n(\vartheta) \pi(d\vartheta) \in]0, 1],$$

so that the associated μ^* -compatibility posterior distribution $\Pi_{\varepsilon}^n \in \mathcal{P}(\Theta)$ takes the form

$$\Pi_{\varepsilon}^n[B] := \frac{1}{C_{\varepsilon}^n} \int_B w_{\varepsilon}^n(\vartheta) \pi(d\vartheta), \quad B \in \mathcal{B}(\Theta). \quad (22)$$

In this representation, the posterior distribution (22) may be interpreted as the prior reweighted by discrepancy-based compatibility scores. Its large-sample behavior is therefore governed by the geometry of the compatibility weights (21), transferring the asymptotic analysis from the sampling level to the induced compatibility function over the parameter space.

The following proposition captures the structural principle underlying the acceptance mechanism. By combining the triangle inequality and symmetry of $\mathcal{T}$ with (12), (15), and (20), it identifies a sufficient condition for pseudo-samples to belong to the acceptance region.

Proposition 3.1. *Let $\varepsilon > 0$ be an admissible excess threshold. Assume that there exists*

$$\Theta_{\varepsilon} \in \mathcal{B}(\Theta)$$

with $\pi[\Theta_{\varepsilon}] > 0$ and such that, for all sufficiently large $n \in \mathbb{N}^$ and π -almost every $\vartheta \in \Theta_{\varepsilon}$, there exist strictly positive coefficients $\alpha_n, \beta_{\vartheta}, \gamma_{\vartheta,n} \in]0, 1[$, where, up to modification on π -null sets,*

$$\vartheta \mapsto \beta_{\vartheta}, \vartheta \mapsto \gamma_{\vartheta,n} \in \mathcal{B}(\Theta_{\varepsilon},]0, 1[),$$

satisfying

$$\alpha_n + \beta_{\vartheta} + \gamma_{\vartheta,n} \leq 1,$$

for which the following two conditions hold.

1. For $n \in \mathbb{N}^*$ large enough and $\mathbf{P}$ -almost every $\omega \in \Omega$,

$$\mathcal{T}_n^\star \leq \varepsilon \alpha_n. \quad (23)$$

2. For π -almost every $\vartheta \in \Theta_\varepsilon$,

$$\mathcal{T}_\vartheta^\star \leq \varepsilon^\star + \varepsilon \beta_\vartheta. \quad (24)$$

Then, for all sufficiently large $n \in \mathbb{N}^*$, $\mathbf{P}$ -almost surely, and for π -almost every $\vartheta \in \Theta_\varepsilon$,

$$\{\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n \mid \mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta) \leq \varepsilon \gamma_{\vartheta,n}\} \subseteq A_{\vartheta,n}(\varepsilon).$$

All asymptotic, almost-everywhere, and coefficient choices appearing in Proposition 3.1 are understood relative to the fixed excess threshold ε . In particular, the index n from which the asymptotic inequalities hold, as well as the corresponding exceptional sets in Ω and Θ , may naturally depend on ε .

The decomposition established in Proposition 3.1 separates the overall acceptance discrepancy into three structurally distinct contributions: an observational fluctuation factor, a model-compatibility profile relative to the minimal asymptotic discrepancy level, and a finite-sample stabilization mechanism associated with the probabilistic representation of the pseudo-sample.

Consistently with (23) and (13), the first coefficient α_n captures the observed variation term $\mathcal{T}_n^\star$, which becomes asymptotically negligible in $\mathbf{P}$ -probability. Under stronger modes of convergence yielding a deterministic asymptotic bound for $\mathcal{T}_n^\star$, the auxiliary coefficient α_n may be chosen explicitly and need not be treated as an independent component of the decomposition.

The bound (24) involving the second coefficient β_ϑ directly reflects the definition (16) of $\varepsilon^\star$ as the minimal asymptotically achievable discrepancy level within the model class. Moreover, this inequality is automatically satisfied for every $\vartheta \in \Theta^\star$, regardless of the specific value of β_ϑ (see (17)).

Finally, the third coefficient $\gamma_{\vartheta,n}$ controls the finite-sample oscillation of the probabilistic representation $\nu_{\vartheta,n}$ around its asymptotic counterpart ν_ϑ . Requiring, for π -almost every $\vartheta \in \Theta_\varepsilon$, the sequence $\{\gamma_{\vartheta,n}\}_{n \in \mathbb{N}^*}$ to be asymptotically bounded away from zero, that is,

$$\bar{\gamma}_\vartheta := \liminf_{n \rightarrow \infty} \gamma_{\vartheta,n} > 0, \quad (25)$$

does not preclude the discrepancy from vanishing asymptotically; it merely prevents the fraction of the acceptance tolerance allocated to finite-sample fluctuations from degenerating.

This discussion motivates the introduction, for every excess threshold ε , of the asymptotic compatibility functional $\Gamma_\varepsilon: \Theta \rightarrow [0, 1]$ defined, for π -almost every $\vartheta \in \Theta$, by

$$\Gamma_\varepsilon(\vartheta) := \liminf_{n \rightarrow \infty} w_\varepsilon^n(\vartheta). \quad (26)$$

According to (21), the functional (26) quantifies the asymptotic persistence of compatibility between the parameter ϑ and the observed data under the acceptance mechanism induced by the threshold ε .

As an immediate consequence of Proposition 3.1, one obtains the following lower estimate for (26).

Corollary 3.1. *Let $\varepsilon > 0$ be an excess threshold and suppose that the assumptions of Proposition 3.1 are satisfied. Then, for π -almost every $\vartheta \in \Theta_\varepsilon$,*

$$\Gamma_\varepsilon(\vartheta) \geq \liminf_{n \rightarrow \infty} \mu_\vartheta^n[\{\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n \mid \mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta) \leq \varepsilon \gamma_{\vartheta,n}\}]. \quad (27)$$

Corollary 3.1 reveals that the asymptotic compatibility quantified by the functional (26) is directly induced by the finite-sample stabilization mechanism governed by the discrepancy $\mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta)$, whenever $\nu_{\vartheta,n}$ remains sufficiently concentrated around ν_ϑ under the probability measure μ_ϑ^n .

3.3 Compatibility persistence and asymptotic acceptance

To formalize asymptotic compatibility under a fixed excess threshold ε , we first introduce two structural assumptions governing the persistence of acceptance in the discrepancy-driven large-sample regime.

The first condition formalizes the idea that the discrepancy functional (10) furnishes a sufficient asymptotic compatibility control for the finite-sample acceptance mechanism. More precisely, we assume here that there exists a measurable subset

$$\Theta_\varepsilon^w \in \mathcal{B}(\Theta)$$

of strictly positive π -mass such that the compatibility weights introduced in (21) satisfy, for all sufficiently large $n \in \mathbb{N}^*$, every $\mathbf{y}^{1:n} \in \mathcal{Y}^n$, and π -almost every $\vartheta \in \Theta_\varepsilon^w$,

$$w_\varepsilon^n(\vartheta) \leq \mu_\vartheta^n[D_{\varepsilon^* + \varepsilon}(\mathbf{y}^{1:n})]. \quad (28)$$

A particularly simple sufficient condition for the relation (28) is that

$$\mathcal{D}(\mathbf{y}^{1:n}, \mathbf{y}_\vartheta^{1:n}) \leq \Delta_{\vartheta,n}(\mathbf{y}_\vartheta^{1:n})$$

for every $\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n$, throughout the admissible range of the quantities involved.

The previous assumption is not intended to impose full posterior consistency or asymptotic identification. It only requires that the events controlled by $\Delta_{\vartheta,n}$ provide a probabilistic lower bound for the acceptance event associated with the shifted threshold $\varepsilon^* + \varepsilon$. Thus, it is satisfied whenever this map is chosen as an upper envelope of the operational discrepancy used by the ABC mechanism.

The second requirement, in connection with the lower estimate (27) of Corollary 3.1, quantifies the asymptotic persistence of the probabilistic representation associated with the pseudo-sample underlying the acceptance mechanism: we now assume the existence of a residual fluctuation level

$$\delta \in [0, 1[$$

and a measurable subset

$$\Theta_\varepsilon^\delta \in \mathcal{B}(\Theta)$$

of strictly positive π -mass such that, for π -almost every $\vartheta \in \Theta_\varepsilon^\delta$ and every $\lambda \in]0, 1[$,

$$\limsup_{n \rightarrow \infty} \mu_\vartheta^n[\{\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n \mid \mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta) > \varepsilon\lambda\}] \leq \delta. \quad (29)$$

Condition (29) is satisfied, for instance, whenever the $\limsup$ therein reduces to a limit equal to zero, in which case one may simply take $\delta = 0$. This occurs, for instance, whenever, for π -almost every $\vartheta \in \Theta_\varepsilon^\delta$, $\nu_{\vartheta,n}$ converges to ν_ϑ in μ_ϑ^n -probability with respect to $\mathcal{T}$.

This assumption is deliberately formulated at a weaker level of asymptotic stabilization than the convergence-in-probability setting above, which corresponds to the special case $\delta = 0$. Allowing $\delta > 0$ accommodates situations in which a residual fraction of simulated probabilistic representations may fail to stabilize around ν_ϑ , while a non-trivial proportion still remains asymptotically compatible.

In ABC settings based on empirical summaries, empirical measures, or Wasserstein-type discrepancies, these assumptions are consistent with the law of large numbers for the samples; see, e.g., [30, 4, 15]. When the representation satisfies this convergence property under μ_ϑ^n for π -almost every $\vartheta \in \Theta_\varepsilon^\delta$, the second assumption holds with $\delta = 0$. When convergence is partial, non-uniform, or affected by simulation noise, the same condition may still hold for some $\delta > 0$. This is the level of generality needed: the forthcoming results do not require collapse of the simulated mechanism, but only a positive asymptotic mass of pseudo-samples whose representations remain close to their reference law.

The following result establishes a form of asymptotic persistence of compatible posterior mass: under the two previous assumptions and the decomposition of Proposition 3.1, asymptotic compatibility does not necessarily imply concentration of the ABC posterior distribution around the minimizers of the limiting discrepancy function. Instead, the discrepancy-driven acceptance mechanism may asymptotically retain a strictly positive fraction of posterior mass over parameters whose limiting discrepancy remains strictly above, yet sufficiently close to, the minimal compatibility level.

Theorem 3.1. *Let $\varepsilon > 0$ be an excess threshold and suppose that the assumptions of Proposition 3.1, together with conditions (25), (28), and (29), are satisfied. Then*

$$\liminf_{n \rightarrow \infty} \pi_{\varepsilon^* + \varepsilon}^n \left[\left\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta \mid \mathcal{T}_\vartheta^* > \varepsilon^* \right\} \right] \geq (1 - \delta) \pi \left[\left\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta \mid \varepsilon^* < \mathcal{T}_\vartheta^* \leq \varepsilon^* + \varepsilon \beta_\vartheta \right\} \right]. \quad (30)$$

In particular, whenever the set on the right-hand side of (30) has strictly positive π -mass, the left-hand side is strictly positive.

Proof. The proof essentially combines the decomposition framework of Proposition 3.1, the lower bound (27) from Corollary 3.1, and the asymptotic conditions (25) and (29).

We first pass from the finite-sample posterior weights to their asymptotic lower envelope. Specifically, applying Fatou's lemma to the representation (3), then restricting the parameter domain to the non-minimal compatibility region selected by (24), and finally invoking (27) in conjunction with (28), we derive

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \pi_{\varepsilon^* + \varepsilon}^n \left[\left\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta \mid \mathcal{T}_\vartheta^* > \varepsilon^* \right\} \right] \\ & \geq \int_{\left\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta \mid \mathcal{T}_\vartheta^* > \varepsilon^* \right\}} \liminf_{n \rightarrow \infty} \mu_\vartheta^n \left[D_{\varepsilon^* + \varepsilon}(\mathbf{y}^{1:n}) \right] \pi(d\vartheta) \\ & \geq \int_{\left\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta \mid \varepsilon^* < \mathcal{T}_\vartheta^* \leq \varepsilon^* + \varepsilon \beta_\vartheta \right\}} \liminf_{n \rightarrow \infty} \mu_\vartheta^n \left[D_{\varepsilon^* + \varepsilon}(\mathbf{y}^{1:n}) \right] \pi(d\vartheta) \\ & \geq \int_{\left\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta \mid \varepsilon^* < \mathcal{T}_\vartheta^* \leq \varepsilon^* + \varepsilon \beta_\vartheta \right\}} \Gamma_\varepsilon(\vartheta) \pi(d\vartheta) \\ & \geq \int_{\left\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta \mid \varepsilon^* < \mathcal{T}_\vartheta^* \leq \varepsilon^* + \varepsilon \beta_\vartheta \right\}} \liminf_{n \rightarrow \infty} \mu_\vartheta^n \left[\left\{ \mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n \mid \mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta) \leq \varepsilon \gamma_{\vartheta,n} \right\} \right] \pi(d\vartheta). \end{aligned}$$

The claim (30) then follows from conditions (25) and (29) once one selects, for π -almost every $\vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta$, an arbitrary intermediate level

$$\lambda \equiv \lambda_\vartheta \in]0, \bar{\gamma}_\vartheta[.$$

For such parameters ϑ , one has $\varepsilon \gamma_{\vartheta,n} > \varepsilon \lambda_\vartheta$ for all sufficiently large $n \in \mathbb{N}^*$, and therefore

$$\begin{aligned} & \liminf_{n \rightarrow \infty} \mu_\vartheta^n \left[\left\{ \mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n \mid \mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta) \leq \varepsilon \gamma_{\vartheta,n} \right\} \right] \\ & = 1 - \limsup_{n \rightarrow \infty} \mu_\vartheta^n \left[\left\{ \mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n \mid \mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta) > \varepsilon \gamma_{\vartheta,n} \right\} \right] \\ & \geq 1 - \limsup_{n \rightarrow \infty} \mu_\vartheta^n \left[\left\{ \mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n \mid \mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta) > \varepsilon \lambda_\vartheta \right\} \right] \geq 1 - \delta, \end{aligned}$$

thereby concluding the argument. $\square$

A sufficient condition for the admissible discrepancy layer in the lower estimate (30) not to collapse onto the minimal compatibility level ε^* is that the coefficient β_ϑ remain uniformly bounded away from zero on $\Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta$, thus preventing the lower bound from becoming asymptotically vacuous.

The compatibility-induced posterior distribution Π_ε^n introduced in (22) inherits an entirely analogous asymptotic persistence principle. In this case, the corresponding derivation becomes more direct, since Π_ε^n is already represented in terms of the compatibility weights w_ε^n .

Proposition 3.2. *Let $\varepsilon > 0$ be an excess threshold and suppose that the assumptions of Proposition 3.1, together with conditions (25) and (29), are satisfied. Then*

$$\liminf_{n \rightarrow \infty} \Pi_\varepsilon^n \left[\left\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^\delta \mid \mathcal{T}_\vartheta^* > \varepsilon^* \right\} \right] \geq (1 - \delta) \pi \left[\left\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^\delta \mid \varepsilon^* < \mathcal{T}_\vartheta^* \leq \varepsilon^* + \varepsilon \beta_\vartheta \right\} \right].$$

It is worth observing that the measures π_ε^n and Π_ε^n need not be comparable in general. Indeed, even if relation (28) were to hold π -almost everywhere with respect to ϑ , the associated normalization procedures would still differ. This reflects the intrinsically distinct nature of the two posterior constructions.

Taken together, the previous results show that the persistence of posterior mass over compatible yet non-minimal regions is not a pathological feature of the asymptotic regime, but rather a structural consequence of discrepancy-based compatibility selection under fixed tolerance levels. In particular, asymptotic compatibility and asymptotic concentration need not coincide in general.

Remark 3.3. Within the present setting, suppose that, for all sufficiently large $n \in \mathbb{N}^*$ and π -almost every

$$\vartheta \in \Theta_\varepsilon^w \cup \Theta_\varepsilon^\delta,$$

the probability measure μ_ϑ^n admits a density $f_\vartheta^n = d\mu_\vartheta^n/d\mathbf{Vol}_{nm_y}$ with respect to the ambient Lebesgue measure, as in Subsection 2.2. Then the structural assumptions (28) and (29) appearing in Theorem 3.1 admit the following respective integral formulations over the corresponding parameter sets:

$$\begin{aligned} \int_{A_{\vartheta,n}(\varepsilon)} f_\vartheta^n(z^{1:n}) dz^{1:n} &\leq \int_{D_{\varepsilon^*+\varepsilon}(\mathbf{y}^{1:n})} f_\vartheta^n(z^{1:n}) dz^{1:n}, \\ \limsup_{n \rightarrow \infty} \int_{\{\mathbf{y}_\vartheta^{1:n} \in \mathcal{Y}_\vartheta^n \mid \mathcal{T}(\nu_{\vartheta,n}, \nu_\vartheta) > \varepsilon\lambda\}} f_\vartheta^n(z^{1:n}) dz^{1:n} &\leq \delta. \end{aligned}$$

Consequently, in the classical density-based formulation of ABC inference, the persistence assumptions above may be ensured through various natural regularity conditions, either on the acceptance regions or directly on the associated family of densities $\{f_\vartheta^n\}_{n \in \mathbb{N}^*}$, over the admissible parameter range.

The previous persistence results admit a direct geometric interpretation at the level of the parameter space. Under a local compatibility-growth condition around a minimizer $\vartheta^* \in \Theta^*$ of the asymptotic discrepancy profile (15) (see (17)), asymptotic persistence of posterior mass may still occur outside arbitrarily small d_Θ -neighborhoods of ϑ^* itself, despite its optimality.

In particular, sufficiently flat local discrepancy functions near compatible minimizers may prevent asymptotic posterior concentration around compatibility-optimal configurations.

Proposition 3.3. *Let $\varepsilon > 0$ be an excess threshold and suppose that the assumptions of Proposition 3.1, together with conditions (25), (28), and (29), are satisfied. Let $\vartheta^* \in \Theta^*$ and assume that there exist a non-degenerate interval $I_{\vartheta^*} \subseteq \mathbb{R}_+$ containing zero, a strictly increasing function*

$$\psi_{\vartheta^*}: I_{\vartheta^*} \longrightarrow \mathbb{R}_+ \quad (31)$$

with $\psi_{\vartheta^}(0) = 0$, and a d_Θ -neighborhood $\Theta_{\vartheta^*} \in \mathcal{B}(\Theta)$ of ϑ^* such that, for π -almost every $\vartheta \in \Theta_{\vartheta^*}$,*

$$\mathcal{T}_\vartheta^* - \varepsilon^* \leq \psi_{\vartheta^*}(d_\Theta(\vartheta, \vartheta^*)). \quad (32)$$

Then, for every $s \in I_{\vartheta^} \setminus \{0\}$, there exists $r_s \in]0, s]$, such that, for every $r \in]0, r_s]$,*

$$\begin{aligned} \liminf_{n \rightarrow \infty} \pi_{\varepsilon^*+\varepsilon}^n[\{\vartheta \in \Theta \mid d_\Theta(\vartheta, \vartheta^*) \geq r\}] \\ \geq (1 - \delta) \pi[\{\vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta \mid \varepsilon^* + \psi_{\vartheta^*}(s) \leq \mathcal{T}_\vartheta^* \leq \varepsilon^* + \varepsilon\beta_\vartheta\}]. \end{aligned} \quad (33)$$

Proof. Fix $s \in I_{\vartheta^*} \setminus \{0\}$. By the d_Θ -neighborhood assumption, we may then choose a sufficiently small radius $r_s \in]0, s]$ so that, for every $r \in]0, r_s]$,

$$B_r(\vartheta^*) := \{\vartheta \in \Theta \mid d_\Theta(\vartheta, \vartheta^*) < r\} \subseteq \Theta_{\vartheta^*}.$$

Combining the compatibility-growth condition (32) with the strict monotonicity of ψ_{ϑ^*} , one then obtains, for π -almost every $\vartheta \in B_r(\vartheta^*)$,

$$\mathcal{T}_\vartheta^* - \varepsilon^* \leq \psi_{\vartheta^*}(d_\Theta(\vartheta, \vartheta^*)) < \psi_{\vartheta^*}(r) \leq \psi_{\vartheta^*}(s).$$

Equivalently, up to a suitable π -null set $\mathcal{N} \in \mathcal{B}(\Theta)$ (depending on the parameters involved),

$$B_r(\vartheta^\star) \setminus \mathcal{N} \subseteq \{ \vartheta \in \Theta \mid \mathcal{T}_\vartheta^\star < \varepsilon^\star + \psi_{\vartheta^\star}(s) \}$$

or, in complementary form,

$$\{ \vartheta \in \Theta \mid \mathcal{T}_\vartheta^\star \geq \varepsilon^\star + \psi_{\vartheta^\star}(s) \} \subseteq \{ \vartheta \in \Theta \mid d_\Theta(\vartheta, \vartheta^\star) \geq r \} \cup \mathcal{N}.$$

Consequently, for every $n \in \mathbb{N}^*$, since the posterior measure $\pi_{\varepsilon^\star + \varepsilon}^n$ is by construction absolutely continuous with respect to π , it assigns zero mass to $\mathcal{N}$ as well, and therefore

$$\begin{aligned} & \pi_{\varepsilon^\star + \varepsilon}^n[\{ \vartheta \in \Theta \mid d_\Theta(\vartheta, \vartheta^\star) \geq r \}] \\ & \geq \pi_{\varepsilon^\star + \varepsilon}^n[\{ \vartheta \in \Theta \mid \mathcal{T}_\vartheta^\star \geq \varepsilon^\star + \psi_{\vartheta^\star}(s) \}] \\ & \geq \pi_{\varepsilon^\star + \varepsilon}^n[\{ \vartheta \in \Theta_\varepsilon \cap \Theta_\varepsilon^w \cap \Theta_\varepsilon^\delta \mid \varepsilon^\star + \psi_{\vartheta^\star}(s) \leq \mathcal{T}_\vartheta^\star \leq \varepsilon^\star + \varepsilon\beta_\vartheta \}]. \end{aligned} \quad (34)$$

Passing now to the lower limit as $n \rightarrow \infty$, we may reproduce the argument developed in the proof of Theorem 3.1 over the restricted measurable set appearing in the last line of (34), thus establishing (33). $\square$

An analogous argument yields the geometric persistence estimate for the compatibility-induced posterior distribution Π_ε^n , with condition (28)—and hence the set Θ_ε^w —no longer needed.

In both instances, geometrically regular situations arise whenever the compatibility-growth profile (31) admits an explicit nondegenerate quantitative form (e.g. of polynomial or Hölder type), while the most regular regime occurs when a common global modulus

$$\psi \equiv \psi_{\Theta^\star}$$

may be selected uniformly over the entire minimizer set $\Theta^\star$.

In this latter situation, the lower estimate in (33) may also be readily reformulated in terms of the distance function to the set $\Theta^\star$, defined by

$$d_{\Theta^\star}(\vartheta) := \inf_{\vartheta^\star \in \Theta^\star} d_\Theta(\vartheta, \vartheta^\star), \quad \vartheta \in \Theta,$$

leading, for sufficiently small $r > 0$, to analogous persistence estimates over tubular-type events of the form

$$\{ \vartheta \in \Theta \mid d_{\Theta^\star}(\vartheta) \geq r \}.$$

This viewpoint highlights the geometric structure underlying persistence estimates well beyond the d_Θ -neighborhoods of individual minimizers $\vartheta^\star \in \Theta^\star$.

These observations indicate that compatibility regions may remain genuinely extended whenever the discrepancy geometry fails to induce sufficiently strong local separation. In particular, persistence phenomena may be viewed as manifestations of the intrinsic geometric structure of the limiting discrepancy function itself, rather than consequences of the failure of asymptotic compatibility.

Remark 3.4. In light of the foregoing, the asymptotic role of ABC posterior distributions is better understood as that of compatibility filters rather than mechanisms for posterior concentration.

Indeed, whenever the acceptance weights converge π -almost everywhere, i.e.,

$$w_\varepsilon^n(\vartheta) \xrightarrow{n \rightarrow \infty} \Gamma_\varepsilon(\vartheta)$$

for π -almost every $\vartheta \in \Theta$, one formally obtains a limiting filtered posterior measure on $\mathcal{B}(\Theta)$ given by

$$\Pi_\varepsilon^\Gamma(d\vartheta) := \frac{\Gamma_\varepsilon(\vartheta) \pi(d\vartheta)}{\int_\Theta \Gamma_\varepsilon(\vartheta') \pi(d\vartheta')},$$

provided that Γ_ε is not π -almost surely equal to zero.

From this perspective, asymptotic ABC inference is not necessarily governed by concentration on the optimal parameter set, but rather by the geometry of the induced compatibility region.

The present framework further suggests possible connections with large-deviation and Laplace-type asymptotic principles (see, e.g., [10, 9, 11]). More precisely, in several discrepancy-based settings, acceptance weights may exhibit exponential asymptotic structures of the form

$$w_{\varepsilon}^n(\cdot) \approx e^{-nJ_{\varepsilon}(\cdot)}$$

for some rate functional J_{ε} , thereby inducing Gibbs-type posterior geometries.

A rigorous study of such asymptotic regimes, however, lies beyond the scope of the present work.

4 Variational and transport structures in ABC

Approximate Bayesian Computation should not be merely regarded as an acceptance–rejection mechanism, nor as a likelihood-free approximation of classical Bayesian updating. Once discrepancies between simulated and observed data are encoded into compatibility weights, ABC naturally induces variational and geometric structures on the space of posterior probability measures.

In this section, posterior distributions are first analyzed through variational principles and their asymptotic counterparts, and are then shown to admit a transport-geometric characterization.

Unlike the regime considered in Section 3, we return here to a fixed-data perspective.

4.1 A compatibility-induced variational principle

Throughout the subsection, let $\varepsilon > 0$ be given, and let $w_{\varepsilon} : \Theta \rightarrow \mathbb{R}_+$ be a compatibility weight as introduced in Subsection 2.1. From a variational perspective, rather than viewing w_{ε} solely as a mechanism assigning a compatibility score to each parameter value $\vartheta \in \Theta$ with respect to the observed sample $\mathbf{y}^{1:n} \in \mathcal{Y}^n$, we regard it as defining a data-dependent energy landscape on the parameter space.

Henceforth, we restrict our attention to compatibility weights satisfying, π -a.s.,

$$0 < w_{\varepsilon} \leq 1. \quad (35)$$

Up to a modification on a π -null set, we may assume that (35) holds everywhere on Θ , and introduce the corresponding compatibility loss $\ell_{\varepsilon} \in \mathcal{B}(\Theta, \mathbb{R})$ as

$$\ell_{\varepsilon} := -\log w_{\varepsilon}. \quad (36)$$

This logarithmic transformation converts the multiplicative compatibility scale into an additive one, playing a role analogous to that of negative log-likelihood functionals in classical statistical inference. Larger values of w_{ε} correspond to smaller values of ℓ_{ε} , so that minimizing the compatibility loss is equivalent to maximizing compatibility with the observed sample. Under the present assumptions, the induced compatibility loss is nonnegative, consistent with the classical ABC setting (see also Subsection 2.2). This additive formulation is particularly well suited for variational analysis, as it naturally gives rise to entropy-regularized and risk-based formulations of approximate Bayesian inference.

A further connection arises with loss-based forms of generalized Bayesian updating; see, e.g., [5]. The compatibility loss (36) plays the role of an empirical loss, while the compatibility weights act as likelihood surrogates. The resulting inferential structure is induced directly by the discrepancy geometry relating observed and simulated data, rather than by an explicit probabilistic observation model.

At the level of posterior candidates, this pointwise loss induces the extended-valued functional

$$\varrho \mapsto \int_{\Theta} \ell_{\varepsilon}(\vartheta) \varrho(d\vartheta), \quad \varrho \in \mathcal{P}(\Theta), \quad (37)$$

which represents the compatibility loss averaged with respect to the candidate posterior distribution ϱ . Thus, posterior candidates are no longer evaluated only through the compatibility of individual parameter values, but through the aggregate loss associated with their distribution over the parameter space.

This construction extends to a general variational framework suited for the subsequent developments. It is based on the interplay between a data-fitting functional

$$\mathcal{R}_{\varepsilon} : \mathcal{P}(\Theta) \longrightarrow [0, \infty]$$

and a regularization functional

$$I(\cdot | \pi): \mathcal{P}(\Theta) \longrightarrow [0, \infty],$$

with the former representing the compatibility component associated with the observed sample $\mathbf{y}^{1:n}$, and the latter acting as a prior-based informational regularization term. This yields the variational problem

$$\inf_{\boldsymbol{\varrho} \in \mathcal{P}(\Theta)} \{ \mathcal{R}_\varepsilon(\boldsymbol{\varrho}) + I(\boldsymbol{\varrho} | \pi) \}. \quad (38)$$

By construction, the effective domain of the objective functional $\mathcal{R}_\varepsilon(\cdot) + I(\cdot | \pi)$ in (38) coincides with the set of posterior candidates $\boldsymbol{\varrho} \in \mathcal{P}(\Theta)$ having finite objective value.

In the canonical compatibility-loss setting introduced above, the functional $\mathcal{R}_\varepsilon$ reduces precisely to (37). In this case, it admits a natural interpretation as a compatibility-induced risk functional, thereby motivating its notation. The regularization term then acts to prevent the minimization from collapsing into a purely compatibility-driven concentration on the most compatible regions of the parameter space, while preserving the role of the prior as an informational reference measure.

Among the possible choices for the functional $I(\cdot | \pi)$, the Kullback–Leibler divergence, also known as relative entropy, emerges as a particularly natural candidate. It is defined on $\mathcal{P}(\Theta)$ by

$$\text{KL}(\boldsymbol{\varrho} | \pi) := \begin{cases} \int_{\Theta} \log \left[\frac{d\boldsymbol{\varrho}}{d\pi}(\boldsymbol{\vartheta}) \right] \boldsymbol{\varrho}(d\boldsymbol{\vartheta}), & \text{if } \boldsymbol{\varrho} \ll \pi, \\ \infty, & \text{otherwise.} \end{cases} \quad (39)$$

The relative entropy quantifies the informational divergence of the posterior candidate $\boldsymbol{\varrho}$ from the prior distribution π . Within this framework, it acts as an informational regularization term, discouraging excessive departures from the prior while balancing the influence of the compatibility component.

Within this compatibility-loss setting, and taking the Kullback–Leibler divergence as the regularization functional, the variational problem (38) specializes to

$$\inf_{\substack{\boldsymbol{\varrho} \in \mathcal{P}(\Theta) \\ \boldsymbol{\varrho} \ll \pi}} \left\{ - \int_{\Theta} \log w_\varepsilon(\boldsymbol{\vartheta}) \boldsymbol{\varrho}(d\boldsymbol{\vartheta}) + \text{KL}(\boldsymbol{\varrho} | \pi) \right\}. \quad (40)$$

This formulation corresponds to an entropy-regularized risk minimization problem over $\mathcal{P}(\Theta)$, where the risk is induced by compatibility with the data. The relative entropy term penalizes deviations from the prior distribution, whereas the compatibility-induced risk drives posterior candidates toward regions of the parameter space exhibiting higher levels of compatibility with the observed sample.

The problem (40) admits a variational characterization: the compatibility-weight posterior distribution $\pi_\varepsilon \in \mathcal{P}(\Theta)$ introduced in (4) is the unique minimizer of the objective functional $\mathcal{F}_\varepsilon$ on $\mathcal{P}(\Theta)$ given by

$$\mathcal{F}_\varepsilon(\boldsymbol{\varrho}) := \begin{cases} - \int_{\Theta} \log w_\varepsilon(\boldsymbol{\vartheta}) \boldsymbol{\varrho}(d\boldsymbol{\vartheta}) + \text{KL}(\boldsymbol{\varrho} | \pi), & \text{if } \boldsymbol{\varrho} \ll \pi, \\ \infty, & \text{otherwise.} \end{cases} \quad (41)$$

This representation is a compatibility-based instance of the classical Gibbs–Donsker–Varadhan principle; see, e.g., [10, 9]. Although standard, it is included for completeness, as it provides the fundamental link between compatibility-weight posteriors and entropy-regularized risk minimization.

Proposition 4.1. *Under the standing assumptions of this subsection, the probability measure π_ε , defined as in (4), is the unique minimizer of $\mathcal{F}_\varepsilon$ on $\mathcal{P}(\Theta)$, with minimum value*

$$\min_{\boldsymbol{\varrho} \in \mathcal{P}(\Theta)} \mathcal{F}_\varepsilon(\boldsymbol{\varrho}) \equiv \mathcal{F}_\varepsilon(\pi_\varepsilon) = - \log \int_{\Theta} w_\varepsilon(\boldsymbol{\vartheta}) \pi(d\boldsymbol{\vartheta}). \quad (42)$$

Proof. Let $C_\varepsilon \in]0, 1]$ denote the normalization constant associated with π_ε , that is,

$$C_\varepsilon := \int_{\Theta} w_\varepsilon(\boldsymbol{\vartheta}) \pi(d\boldsymbol{\vartheta}). \quad (43)$$

By construction,

$$\frac{w_\varepsilon}{C_\varepsilon} = \frac{d\pi_\varepsilon}{d\pi},$$

and hence

$$\log w_\varepsilon = \log C_\varepsilon + \log \frac{d\pi_\varepsilon}{d\pi}.$$

Moreover, it follows from assumption (35) that π_ε and π are equivalent measures. Combining (39) and (41), we then obtain, for every $\varrho \in \mathcal{P}(\Theta)$,

$$\mathcal{F}_\varepsilon(\varrho) = -\log C_\varepsilon + \text{KL}(\varrho \mid \pi_\varepsilon). \quad (44)$$

The non-negativity of the Kullback–Leibler divergence term appearing in (44), together with the fact that it vanishes if and only if $\varrho = \pi_\varepsilon$, yields the stated characterization. $\square$

Proposition 4.1 shows that the posterior distribution π_ε embodies more than the mere normalization of compatibility weights. Rather, it emerges as the unique equilibrium probability measure intrinsically determined by an entropy-regularized compatibility principle.

Posterior updating may thus be interpreted as the selection of the probability measure that optimally balances compatibility with the observed data and informational coherence with the prior distribution.

Within this interpretation, the normalization constant (43) plays the role of a partition function, while its negative logarithm represents the minimum entropy-regularized compatibility risk.

At this stage, the natural question is whether this variational structure remains stable under the vanishing-threshold regime $\varepsilon \downarrow 0$ of compatibility weights identified in Proposition 2.1, and thus persists in the asymptotic regime. The next result confirms that this is indeed the case.

Theorem 4.1. *Under the standing assumptions of this subsection and of Proposition 2.1, suppose further that $\bar{w} > 0$ π -a.s., and define, for every $\varrho \in \mathcal{P}(\Theta)$,*

$$\tilde{\mathcal{F}}_\varepsilon(\varrho) := \begin{cases} -\int_\Theta \log \left[\frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) \right] \varrho(d\vartheta) + \text{KL}(\varrho \mid \pi), & \text{if } \varrho \ll \pi, \\ \infty, & \text{otherwise,} \end{cases} \quad (45)$$

and

$$\bar{\mathcal{F}}(\varrho) := -\log \int_\Theta \bar{w}(\vartheta) \pi(d\vartheta) + \text{KL}(\varrho \mid \bar{\pi}), \quad (46)$$

where $\bar{\pi}$ is defined in (6). Then π_ε is the unique minimizer of $\tilde{\mathcal{F}}_\varepsilon$ on $\mathcal{P}(\Theta)$, with minimum value

$$\min_{\varrho \in \mathcal{P}(\Theta)} \tilde{\mathcal{F}}_\varepsilon(\varrho) \equiv \tilde{\mathcal{F}}_\varepsilon(\pi_\varepsilon) = -\log \int_\Theta \frac{1}{\kappa_\varepsilon} w_\varepsilon(\vartheta) \pi(d\vartheta). \quad (47)$$

Likewise, the probability measure $\bar{\pi}$ is the unique minimizer of $\bar{\mathcal{F}}$ on $\mathcal{P}(\Theta)$, and

$$\min_{\varrho \in \mathcal{P}(\Theta)} \bar{\mathcal{F}}(\varrho) \equiv \bar{\mathcal{F}}(\bar{\pi}) = -\log \int_\Theta \bar{w}(\vartheta) \pi(d\vartheta). \quad (48)$$

Furthermore,

$$\lim_{\varepsilon \downarrow 0} \tilde{\mathcal{F}}_\varepsilon(\pi_\varepsilon) = \bar{\mathcal{F}}(\bar{\pi}). \quad (49)$$

Finally, if $\{\varrho_\varepsilon\}_{\varepsilon>0}$ is any family in $\mathcal{P}(\Theta)$ of asymptotic minimizers, that is,

$$\lim_{\varepsilon \downarrow 0} \left[\tilde{\mathcal{F}}_\varepsilon(\varrho_\varepsilon) - \tilde{\mathcal{F}}_\varepsilon(\pi_\varepsilon) \right] = 0, \quad (50)$$

then this family converges in total variation to $\bar{\pi}$.

Proof. The characterization of π_ε as the unique minimizer of $\tilde{\mathcal{F}}_\varepsilon$, together with identity (47), is established by the same argument used in the proof of Proposition 4.1, applied to the normalized compatibility weight $\kappa_\varepsilon^{-1} w_\varepsilon$. The corresponding statement for $\bar{\mathcal{F}}$ is immediate from definition (46). Identity (49) then follows by combining (47) and (48) with the $L^1(\pi)$ -convergence (5) as $\varepsilon \downarrow 0$.

It now remains to demonstrate the stability of asymptotic minimizers $\{\varrho_\varepsilon\}_{\varepsilon>0}$. By the variational identity corresponding to $\tilde{\mathcal{F}}_\varepsilon$ and analogous to (44), one has

$$\tilde{\mathcal{F}}_\varepsilon(\varrho_\varepsilon) - \tilde{\mathcal{F}}_\varepsilon(\pi_\varepsilon) = \text{KL}(\varrho_\varepsilon \mid \pi_\varepsilon),$$

and, in particular, condition (50) may be rewritten as

$$\lim_{\varepsilon \downarrow 0} \text{KL}(\varrho_\varepsilon \mid \pi_\varepsilon) = 0. \quad (51)$$

On the other hand, Pinsker's inequality gives

$$\sup_{B \in \mathcal{B}(\Theta)} |\varrho_\varepsilon[B] - \pi_\varepsilon[B]| \leq \sqrt{\frac{1}{2} \text{KL}(\varrho_\varepsilon \mid \pi_\varepsilon)}, \quad (52)$$

which, combined with (51), shows that the left-hand side in (52) also vanishes as $\varepsilon \downarrow 0$.

In conclusion, since Proposition 2.1 ensures that $\{\pi_\varepsilon\}_\varepsilon$ converges to $\bar{\pi}$ in total variation, an application of the triangle inequality implies the same convergence for $\{\varrho_\varepsilon\}_{\varepsilon>0}$. $\square$

Theorem 4.1 carries implications that extend beyond the posterior convergence established in Proposition 2.1. Indeed, the entire variational mechanism persists in the asymptotic regime: the limiting profile induces a variational problem governed by the same structural principle as its finite-threshold counterparts, and its minimizer is precisely the limiting posterior distribution.

From a risk-based perspective, the theorem identifies asymptotic optimality as the governing principle for the limiting statistical behavior of posterior candidates. Any family whose excess entropy-regularized compatibility risk vanishes asymptotically must converge to $\bar{\pi}$. Thus, the limiting compatibility posterior emerges not only as the optimizer of the asymptotic variational problem, but also as the unique statistically stable posterior selected by asymptotic risk minimization.

Remark 4.1. One may naturally interpret Theorem 4.1 through the lens of variational convergence theory, and in particular of Γ -convergence; see, e.g., [8]. From this viewpoint, since the convergence of minimum values and the stability of asymptotic minimizers are characteristic consequences of variational convergence principles, the family of entropy-regularized compatibility functionals $\tilde{\mathcal{F}}_\varepsilon$, $\varepsilon > 0$, may be viewed as asymptotically governed by the limiting functional $\bar{\mathcal{F}}$. This interpretation further suggests that the vanishing-threshold regime should be understood not merely as a convergence of posterior distributions, but more fundamentally as a convergence of the underlying inferential problems themselves.

A rigorous treatment of this connection would require the development of an explicit Γ -convergence theory, a perspective that we leave for future investigation.

The variational stability established in Theorem 4.1 reveals an underlying principle that is not specific to the entropy-based setting considered above. In the present framework, relative entropy defines the informational geometry associated with the prior, while alternative regularizers may induce different notions of statistical complexity, regularity, or proximity.

The asymptotic regime developed in Section 3 provides a transparent realization of this broader perspective. Indeed, whenever the limiting discrepancy profile (15) is available, namely

$$\vartheta \mapsto \mathcal{T}_\vartheta^\star, \quad \vartheta \in \Theta,$$

it induces a canonical asymptotic notion of compatibility on the parameter space, since, for each $\vartheta \in \Theta$, the value $\mathcal{T}_\vartheta^\star$ measures the asymptotic discrepancy between the probabilistic structure generated under ϑ and the limiting observational distribution associated with the data. Hence, smaller values correspond to statistical models exhibiting a higher degree of asymptotic compatibility with the observed sample.

This naturally leads to the asymptotic compatibility functional

$$\mathcal{R}_\varphi^\star(\varrho) := \int_{\Theta} \varphi(\mathcal{T}_\vartheta^\star - \varepsilon^\star) \varrho(d\vartheta), \quad \varrho \in \mathcal{P}(\Theta), \varrho \ll \pi \quad (53)$$

(understood in the extended sense) associated with a prescribed strictly increasing penalty function

$$\varphi: \mathbb{R}_+ \longrightarrow \mathbb{R}_+,$$

continuous at zero and satisfying $\varphi(0) = 0$, and where $\varepsilon^\star$ is the asymptotic threshold (16).

The profile $\vartheta \mapsto \mathcal{T}_\vartheta^\star$ determines the intrinsic asymptotic compatibility structure on Θ , while the transformation φ specifies how deviations from the optimal compatibility level are quantified. The resulting functional $\mathcal{R}_\varphi^\star$ evaluates the average penalized asymptotic incompatibility of posterior candidates, separating the asymptotic discrepancy profile from the penalization mechanism.

Remark 4.2. The functional $\mathcal{R}_\varphi^\star$ extends the asymptotic compatibility structure from Θ to $\mathcal{P}(\Theta)$. It evaluates posterior candidates through their average penalized asymptotic incompatibility.

This interpretation is fully consistent with the threshold $\varepsilon^\star$ and the associated minimal set (17), $\Theta^\star$, introduced in Section 3. Indeed, a straightforward argument shows that

$$\inf_{\substack{\varrho \in \mathcal{P}(\Theta) \\ \varrho \ll \pi}} \mathcal{R}_\varphi^\star(\varrho) = 0,$$

where equality is attained if and only if $\varrho[\Theta^\star] = 1$.

Finally, owing to its integral definition, the functional (53) also admits close connections with the asymptotic persistence results established in Theorem 3.1, Proposition 3.2, and Proposition 3.3. From this standpoint, these persistence phenomena and the variational formulation associated with $\mathcal{R}_\varphi^\star$ may be viewed as complementary expressions of the same underlying compatibility structure, offering probabilistic and variational perspectives on a common asymptotic behavior.

Among the possible regularization strategies, an optimal-transport approach constitutes a particularly coherent extension of the framework developed so far.

As anticipated in Section 3, transport-based discrepancy maps (9) naturally induce compatibility mechanisms in Approximate Bayesian Computation. The same geometric principles may then be used to endow the space $\mathcal{P}(\Theta)$ of posterior candidates with a corresponding transport structure.

These considerations motivate the formulation introduced in the next subsection.

4.2 Optimal transport structures

In this subsection, compatibility is characterized via transport discrepancies on the data space, while regularization is governed by transport geometry on the space of posterior distributions. The analysis is carried out within the standard theory of optimal transport; see, e.g., the foundational contribution [22], the classical monographs [35, 36], and the more recent treatments [2, 32, 31].

The observation space $(\mathcal{Y}, d_{\mathcal{Y}})$ is assumed to be Polish. With separability already ensured, the only additional hypothesis is completeness. This is the case, for instance, when $\mathcal{Y}$ is a closed subset of $\mathbb{R}^{m_{\mathcal{Y}}}$ endowed with the induced Euclidean metric. Besides encompassing most statistical models encountered in practice, this provides the regularity needed for the transport-theoretic constructions.

We restrict attention to the classical Wasserstein setting generated by the ground metric $d_{\mathcal{Y}}$, although several of the constructions below admit extensions to more general transport costs.

To this end, we fix once and for all an exponent

$$p \in [1, \infty[.$$

Notation. Let $\mathcal{P}_p(\mathcal{Y})$ denote the subclass of $\mathcal{P}(\mathcal{Y})$ consisting of all probability measures η on $\mathcal{Y}$ admitting a finite p -th moment; i.e., for some (equivalently, every) $y_0 \in \mathcal{Y}$,

$$\int_{\mathcal{Y}} d_{\mathcal{Y}}^p(y, y_0) \eta(dy) < \infty.$$

We equip $\mathcal{P}_p(\mathcal{Y})$ with the p -Wasserstein distance associated with $d_{\mathcal{Y}}$, defined by

$$W_p(\mu, \nu) := \left[\inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathcal{Y} \times \mathcal{Y}} d_{\mathcal{Y}}^p(y, z) \gamma(dy, dz) \right]^{1/p}, \quad \mu, \nu \in \mathcal{P}_p(\mathcal{Y}),$$

where $\Pi(\mu, \nu)$ denotes the set of all couplings of μ and ν , that is, all probability measures on $\mathcal{Y} \times \mathcal{Y}$ whose first and second marginals are μ and ν , respectively.

The above definition coincides with the Kantorovich formulation of the optimal transport problem corresponding to the cost function $d_{\mathcal{Y}}^p$. Under the present assumptions, this problem admits a minimizer, namely an optimal transport plan. Moreover, the Wasserstein space $(\mathcal{P}_p(\mathcal{Y}), W_p)$ is again Polish.

Accordingly, the framework introduced in Subsection 3.1 is henceforth understood upon replacing $\mathcal{P}(\mathcal{Y})$ with $\mathcal{P}_p(\mathcal{Y})$. Specifically, for every $\mu, \nu \in \mathcal{P}_p(\mathcal{Y})$, we set

$$\mathcal{T}(\mu, \nu) := W_p(\mu, \nu).$$

The statistical model associates with π -almost every parameter value $\vartheta \in \Theta$ the law $\nu_{\vartheta} \in \mathcal{P}_p(\mathcal{Y})$, introduced in (14). The family $\{\nu_{\vartheta}\}_{\vartheta \in \Theta}$ provides a representation of the model class in $\mathcal{P}_p(\mathcal{Y})$, while the limiting measure $\mu^* \in \mathcal{P}_p(\mathcal{Y})$, specified in (11), encodes the observational information.

In particular, the asymptotic compatibility profile (15) is given by

$$\mathcal{T}_{\vartheta}^* = W_p(\mu^*, \nu_{\vartheta}) \tag{54}$$

for π -almost every $\vartheta \in \Theta$, with the asymptotic compatibility threshold (16) being

$$\varepsilon^* = \min_{\vartheta \in \Theta} W_p(\mu^*, \nu_{\vartheta}).$$

The asymptotic compatibility functional (53) then takes the explicit Wasserstein form

$$\mathcal{R}_{\varphi, p}^*(\varrho) := \int_{\Theta} \varphi(W_p(\mu^*, \nu_{\vartheta}) - \varepsilon^*) \varrho(d\vartheta), \quad \varrho \in \mathcal{P}(\Theta), \quad \varrho \ll \pi, \tag{55}$$

where $\varphi: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is the strictly increasing penalty function continuous at zero with $\varphi(0) = 0$.

As established in Remark 4.2, for every $\varrho \in \mathcal{P}(\Theta)$ with $\varrho \ll \pi$,

$$\mathcal{R}_{\varphi, p}^*(\varrho) = 0 \iff \varrho[\Theta^*] = 1,$$

where Θ^* is the asymptotically optimal set introduced in (17).

Altogether, these observations strongly suggest that asymptotic compatibility should be regarded as entirely determined by the model laws generated by the parameters. It is therefore natural to transfer the variational analysis from the parameter space Θ to the Wasserstein space $\mathcal{P}_p(\mathcal{Y})$.

To formalize this idea, we assume that the parameter-to-distribution correspondence $\vartheta \mapsto \nu_{\vartheta}$ admits a Borel measurable version $\Phi: \Theta \rightarrow \mathcal{P}_p(\mathcal{Y})$ such that, for π -almost every $\vartheta \in \Theta$,

$$\Phi(\vartheta) = \nu_{\vartheta}.$$

In this way, every posterior candidate $\varrho \in \mathcal{P}(\Theta)$, $\varrho \ll \pi$, induces the push-forward (or image) measure

$$\Lambda_{\varrho} := \Phi_{\#} \varrho \in \mathcal{P}(\mathcal{P}_p(\mathcal{Y})), \tag{56}$$

which represents the distribution of the random probability measure $\Phi(\vartheta) \equiv \nu_{\vartheta}$ when the parameter ϑ is distributed according to ϱ . Indeed, by definition,

$$\Phi_{\#} \varrho[\mathcal{E}] = \varrho[\Phi^{-1}(\mathcal{E})], \quad \mathcal{E} \in \mathcal{B}(\mathcal{P}_p(\mathcal{Y})).$$

Moreover, it immediately follows that $\Lambda_{\varrho} \ll \Lambda_{\pi}$, so that

$$W_p(\mu^*, \nu) \geq \varepsilon^* \quad \Lambda_{\varrho}\text{-a.s.} \tag{57}$$

Applying the standard integration formula for image measures to (55), we obtain

$$\mathcal{R}_{\varphi,p}^*(\varrho) = \int_{\mathcal{P}_p(\mathcal{Y})} \varphi(W_p(\mu^*, \nu) - \varepsilon^*) \Lambda_{\varrho}(\mathrm{d}\nu), \quad \varrho \in \mathcal{P}(\Theta), \varrho \ll \pi. \quad (58)$$

Hence, posterior candidates are compared exclusively through the distributions of model laws, rather than through the parameters themselves. Statistical information is carried by the transport discrepancy from the observational law μ^* , while ϱ enters only through the push-forward measure Λ_{ϱ} .

In particular, any two probability measures $\varrho, \tilde{\varrho} \in \mathcal{P}(\Theta)$ with $\varrho, \tilde{\varrho} \ll \pi$ are indistinguishable with respect to asymptotic transport compatibility whenever they share the same induced law; namely,

$$\Lambda_{\varrho} = \Lambda_{\tilde{\varrho}} \implies \mathcal{R}_{\varphi,p}^*(\varrho) = \mathcal{R}_{\varphi,p}^*(\tilde{\varrho}).$$

The measure (56) thus serves as the natural transport-geometric representative of each posterior candidate.

Remark 4.3. If φ is lower semicontinuous (i.e., left-continuous), then, by standard arguments, $\mathcal{R}_{\varphi,p}^*$ is lower semicontinuous with respect to weak convergence of the induced measures Λ_{ϱ} . Thus, asymptotic compatibility cannot deteriorate discontinuously under weak perturbations of model-law distributions.

Furthermore, linearity of integration implies that $\mathcal{R}_{\varphi,p}^*$ is affine with respect to posterior mixtures. This reflects the fact that posterior randomization is evaluated by averaging transport-based compatibility.

The preceding reformulation naturally calls for considering the second-order Wasserstein geometry of

$$\mathcal{P}_p(\mathcal{P}_p(\mathcal{Y})). \quad (59)$$

We equip this space with the p -Wasserstein distance $\mathbb{W}_p$ induced by the ground metric W_p on $\mathcal{P}_p(\mathcal{Y})$:

$$\mathbb{W}_p(\Lambda_1, \Lambda_2) := \left[\inf_{\Gamma \in \Pi(\Lambda_1, \Lambda_2)} \int_{\mathcal{P}_p(\mathcal{Y}) \times \mathcal{P}_p(\mathcal{Y})} W_p^p(\mu, \nu) \Gamma(\mathrm{d}\mu, \mathrm{d}\nu) \right]^{1/p}, \quad \Lambda_1, \Lambda_2 \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{Y})).$$

This lifted geometry admits a natural stability estimate for asymptotic compatibility, offering a first indication of its suitability within the present variational framework. Specifically, whenever the penalty function is Lipschitz continuous, the associated asymptotic compatibility functional inherits the same regularity with respect to the second-order p -Wasserstein metric between the induced laws. An analogous conclusion can be drawn for Hölder-continuous penalty functions.

Proposition 4.2. *Under the standing assumptions of this subsection, suppose that φ is L_{φ} -Lipschitz continuous for some $L_{\varphi} \in]0, \infty[$. Then, for every $\varrho, \tilde{\varrho} \in \mathcal{P}(\Theta)$ with $\varrho, \tilde{\varrho} \ll \pi$ and*

$$\Lambda_{\varrho}, \Lambda_{\tilde{\varrho}} \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{Y})), \quad (60)$$

one has

$$|\mathcal{R}_{\varphi,p}^*(\varrho) - \mathcal{R}_{\varphi,p}^*(\tilde{\varrho})| \leq L_{\varphi} \mathbb{W}_p(\Lambda_{\varrho}, \Lambda_{\tilde{\varrho}}).$$

Proof. For any $\Gamma \in \Pi(\Lambda_{\varrho}, \Lambda_{\tilde{\varrho}})$, by combining (58), the L_{φ} -Lipschitz continuity of φ , the (reverse) triangle inequality for W_p , and Hölder's inequality, we deduce

$$\begin{aligned} |\mathcal{R}_{\varphi,p}^*(\varrho) - \mathcal{R}_{\varphi,p}^*(\tilde{\varrho})| &\leq L_{\varphi} \int_{\mathcal{P}_p(\mathcal{Y}) \times \mathcal{P}_p(\mathcal{Y})} W_p(\mu, \nu) \Gamma(\mathrm{d}\mu, \mathrm{d}\nu) \\ &\leq L_{\varphi} \left[\int_{\mathcal{P}_p(\mathcal{Y}) \times \mathcal{P}_p(\mathcal{Y})} W_p^p(\mu, \nu) \Gamma(\mathrm{d}\mu, \mathrm{d}\nu) \right]^{1/p}. \end{aligned}$$

Since Γ was arbitrary, taking the infimum over all couplings in $\Pi(\Lambda_{\varrho}, \Lambda_{\tilde{\varrho}})$ yields the desired estimate. $\square$

Remark 4.4. The p -moment requirement (60) is fairly mild. In particular, whenever it holds for the prior π , it remains valid for every posterior whose density with respect to π is essentially bounded. More generally, this property follows under suitable Hölder-type conditions on the Radon–Nikodym derivative.

A particularly transparent interpretation emerges when the minimal compatibility threshold vanishes, under a distinguished power-type specialization. In this case, the transport-induced compatibility functional is exactly the p -th power of the second-order p -Wasserstein distance between the Dirac mass concentrated at the observational law and the induced law of model distributions.

Proposition 4.3. *Under the standing assumptions of this subsection, suppose that*

$$\varepsilon^\star = 0$$

and

$$\varphi(x) = x^p, \quad x \in \mathbb{R}_+.$$

Then, for every $\varrho \in \mathcal{P}(\Theta)$ with $\varrho \ll \pi$ and $\Lambda_\varrho \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{Y}))$,

$$\mathcal{R}_{\varphi,p}^\star(\varrho) = \mathbb{W}_p^p(\delta_{\mu^\star}, \Lambda_\varrho).$$

Proof. Let $\varrho \in \mathcal{P}(\Theta)$, $\varrho \ll \pi$, satisfy $\Lambda_\varrho \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{Y}))$. Since the only coupling between $\delta_{\mu^\star}$ and Λ_ϱ is

$$\delta_{\mu^\star} \otimes \Lambda_\varrho,$$

we directly obtain

$$\mathbb{W}_p^p(\delta_{\mu^\star}, \Lambda_\varrho) = \int_{\mathcal{P}_p(\mathcal{Y})} W_p^p(\mu^\star, \nu) \Lambda_\varrho(d\nu) \equiv \mathcal{R}_{\varphi,p}^\star(\varrho),$$

as asserted. $\square$

Proposition 4.3 identifies the degree of asymptotic compatibility of a posterior distribution with the p -th power of the second-order p -Wasserstein distance between $\delta_{\mu^\star}$ and its transport-geometric representative. In this sense, asymptotic compatibility is governed by the geometry of the space (59).

The same geometric principle extends naturally to arbitrary compatibility thresholds, where the reference Dirac mass is replaced by the canonical closed Wasserstein ball.

With this aim, let

$$\bar{B}_{\varepsilon^\star}(\mu^\star) := \{\eta \in \mathcal{P}_p(\mathcal{Y}) \mid W_p(\mu^\star, \eta) \leq \varepsilon^\star\}, \quad (61)$$

and let $d_{\bar{B}_{\varepsilon^\star}(\mu^\star)}: \mathcal{P}_p(\mathcal{Y}) \rightarrow \mathbb{R}_+$ be the associated distance function, given by

$$d_{\bar{B}_{\varepsilon^\star}(\mu^\star)}(\cdot) := \inf_{\eta \in \bar{B}_{\varepsilon^\star}(\mu^\star)} W_p(\cdot, \eta). \quad (62)$$

Theorem 4.2. *Under the standing assumptions of this subsection, suppose that the space $(\mathcal{P}_p(\mathcal{Y}), W_p)$ is geodesic and that $\varphi(x) = x^p$, $x \in \mathbb{R}_+$. Then, for every $\varrho \in \mathcal{P}(\Theta)$ with $\varrho \ll \pi$,*

$$\mathcal{R}_{\varphi,p}^\star(\varrho) = \int_{\mathcal{P}_p(\mathcal{Y})} d_{\bar{B}_{\varepsilon^\star}(\mu^\star)}^p(\nu) \Lambda_\varrho(d\nu).$$

Proof. In view of (57) and (58), it suffices to prove that, for every $\nu \in \mathcal{P}_p(\mathcal{Y})$,

$$d_{\bar{B}_{\varepsilon^\star}(\mu^\star)}(\nu) = (W_p(\mu^\star, \nu) - \varepsilon^\star)_+, \quad (63)$$

where $(\cdot)_+$ denotes the positive-part operator on $\mathbb{R}$.

The inequality “ $\geq$ ” in (63) is an immediate consequence of the metric structure of $\mathcal{P}_p(\mathcal{Y})$. Indeed, by the triangle inequality for W_p , for every $\nu \in \mathcal{P}_p(\mathcal{Y})$ and $\eta \in \bar{B}_{\varepsilon^\star}(\mu^\star)$ (see (61)),

$$W_p(\nu, \eta) \geq W_p(\mu^\star, \nu) - W_p(\mu^\star, \eta) \geq W_p(\mu^\star, \nu) - \varepsilon^\star,$$

whence the claim follows by (62).

The converse inequality is precisely where the assumed geodesic structure of $\mathcal{P}_p(\mathcal{Y})$ comes into play. For this purpose, fix $\nu \in \mathcal{P}_p(\mathcal{Y})$ with $W_p(\mu^\star, \nu) > \varepsilon^\star$, set

$$\sigma_\nu^\star := \frac{\varepsilon^\star}{W_p(\mu^\star, \nu)} \in [0, 1],$$

and let $(\eta_s)_{s \in [0,1]}$ be the constant-speed Wasserstein geodesic in $\mathcal{P}_p(\mathcal{Y})$ connecting $\mu^\star$ and ν . Then

$$W_p(\mu^\star, \eta_{\sigma_\nu^\star}) = \sigma_\nu^\star W_p(\mu^\star, \nu) \equiv \varepsilon^\star$$

and

$$W_p(\nu, \eta_{\sigma_\nu^\star}) = (1 - \sigma_\nu^\star) W_p(\mu^\star, \nu) \equiv W_p(\mu^\star, \nu) - \varepsilon^\star.$$

Therefore, by construction, $\eta_{\sigma_\nu^\star} \in \bar{B}_{\varepsilon^\star}(\mu^\star)$ and

$$d_{\bar{B}_{\varepsilon^\star}(\mu^\star)}(\nu) \leq W_p(\nu, \eta_{\sigma_\nu^\star}) = W_p(\mu^\star, \nu) - \varepsilon^\star.$$

This completes the proof. $\square$

Remark 4.5. The geodesic assumption in Theorem 4.2 holds, for example, whenever the underlying space $(\mathcal{Y}, d_{\mathcal{Y}})$ is geodesic (as the Polish assumption is already in place); see, e.g., [26, 36, 1]. In particular, this includes the classical Euclidean setting and complete Riemannian manifolds.

The preceding results provide a natural transport-geometric characterization of asymptotic compatibility. Posterior candidates are no longer compared directly on the parameter space, but rather through the induced distributions of the model laws in Wasserstein space, while the optimal compatibility threshold $\varepsilon^\star$ determines the reference set against which transport discrepancies are measured. In this sense, the parameter space no longer carries the relevant geometric information.

We conclude by illustrating the preceding constructions through several classes of statistical models. The transport representations in the first two examples are classical; see, e.g., [36, 32].

Example 4.1 (Gaussian model laws). Let $\mathcal{Y} = \mathbb{R}^{m_{\mathcal{Y}}}$ with its Euclidean metric, and suppose that

$$\mu^\star = \mathcal{N}(M^\star, \Sigma^\star)$$

and, for π -almost every $\vartheta \in \Theta$,

$$\nu_\vartheta = \mathcal{N}(M_\vartheta, \Sigma_\vartheta),$$

where $M_\vartheta, M^\star \in \mathbb{R}^{m_{\mathcal{Y}}}$ are mean vectors and $\Sigma_\vartheta, \Sigma^\star \in \mathbb{R}^{m_{\mathcal{Y}} \times m_{\mathcal{Y}}}$ are covariance matrices. Then, for $p = 2$,

$$W_2^2(\mu^\star, \nu_\vartheta) = \|M^\star - M_\vartheta\|^2 + \text{tr} \left[\Sigma^\star + \Sigma_\vartheta - 2 \left[(\Sigma^\star)^{1/2} \Sigma_\vartheta (\Sigma^\star)^{1/2} \right]^{1/2} \right]$$

(Bures–Wasserstein formula; see, e.g., [19]). Consequently, both the asymptotic compatibility profile (54) and the integrand of the associated transport-based compatibility functional (55) admit explicit expressions in terms of the corresponding mean vectors and covariance matrices.

Example 4.2 (Finite-state models). Consider the case where $\mathcal{Y} = \{y_1, \dots, y_m\} \subset \mathbb{R}^{m_{\mathcal{Y}}}$, $m \in \mathbb{N}^*$, and let

$$\mu^\star = \sum_{i=1}^m q_i^\star \delta_{y_i}$$

and, for π -almost every $\vartheta \in \Theta$,

$$\nu_\vartheta = \sum_{i=1}^m q_{\vartheta,i} \delta_{y_i},$$

where $q_i^\star$ and $q_{\vartheta,i}$, $i = 1, \dots, m$, denote the corresponding probability weights. Then

$$W_p^p(\mu^\star, \nu_\vartheta) = \min_{\substack{\gamma_{ij} \geq 0 \\ \sum_{j=1}^m \gamma_{ij} = q_i^\star \\ \sum_{i=1}^m \gamma_{ij} = q_{\vartheta,j}}} \sum_{i,j=1}^m \gamma_{ij} d_{\mathcal{Y}}^p(y_i, y_j).$$

Hence, the asymptotic compatibility profile is determined by finite-dimensional optimal transport problems, and the associated functional by their posterior average, covering categorical and finite-state models.

The formulation illustrated by Examples 4.1 and 4.2 also applies to mixed discrete–continuous laws. For instance, a zero-inflated model may take the form

$$\nu_{\vartheta} = a_{\vartheta} \delta_{y_0} + (1 - a_{\vartheta}) \eta_{\vartheta}, \quad a_{\vartheta} \in [0, 1],$$

where $y_0 \in \mathcal{Y}$, and $\eta_{\vartheta} \in \mathcal{P}_p(\mathcal{Y})$ may be, for example, a Gaussian distribution or a finite Gaussian mixture. The corresponding Wasserstein discrepancy is defined directly between the resulting hybrid probability measures, without requiring separate transport criteria for their discrete and continuous components.

Deterministic models are recovered as an immediate special case. Indeed, if

$$\mu^{\star} = \delta_{y^{\star}}$$

and, for π -almost every $\vartheta \in \Theta$,

$$\nu_{\vartheta} = \delta_{y_{\vartheta}},$$

where $y^{\star}, y_{\vartheta} \in \mathcal{Y}$, then

$$W_p(\mu^{\star}, \nu_{\vartheta}) = d_{\mathcal{Y}}(y^{\star}, y_{\vartheta}).$$

Therefore, in the deterministic setting, the Wasserstein discrepancy reduces exactly to the underlying distance between the observational and model outputs.

5 Conclusions

In this paper, we developed a unified mathematical framework for discrepancy-based Bayesian inference, formulated in terms of compatibility between observed data and the statistical model. Compatibility weights provide a measure-theoretic construction of posterior distributions, while the vanishing-threshold and large-sample regimes describe two distinct asymptotic aspects of the inferential procedure. This formulation also reveals natural variational, transport, and risk-theoretic structures connecting posterior selection, Wasserstein geometry, and compatibility-based assessments of statistical discrepancy.

The main conceptual conclusion is that compatibility is not merely a computational surrogate for an intractable likelihood. Rather, it constitutes the common mathematical object through which posterior construction, asymptotic behavior, variational principles, and transport geometry can be studied within a unified setting. In particular, compatibility weights induce Gibbs-type posterior structures through entropy-regularized variational principles, while transport discrepancies provide an intrinsic representation of the large-sample compatibility profile. These two perspectives are complementary manifestations of the same inferential structure, placing discrepancy-based Bayesian inference at the intersection of likelihood-free inference, generalized Bayesian updating, and optimal transport; see, e.g., [4, 5].

The asymptotic analysis emphasizes that posterior concentration is not an automatic consequence of increasing sample size. Its occurrence depends on the relation between the limiting observational law, the family of sampling distributions, and the compatibility threshold determined by the discrepancy. The minimal compatibility set need not reduce to a singleton, particularly under model misspecification or insufficient identifiability. From this perspective, the transport discrepancy goes beyond quantifying approximation error: it determines the asymptotic geometry of the inferential problem and identifies the probabilistic models that remain asymptotically compatible with the limiting observational law.

Natural directions for further research include more general classes of discrepancies, compatibility weights, and transport costs, as well as finer forms of asymptotic stability and concentration. It may also be extended to structured or infinite-dimensional observations, conditional and sequential models, and statistical experiments in which the observational law is itself subject to uncertainty. More broadly, the variational, transport, and risk-theoretic perspectives may foster compatibility-based posterior procedures and further connections with distributionally robust optimization over Wasserstein ambiguity sets and multivariate risk assessment, as well as with robust Bayesian inference; see, e.g., [14, 18, 25].

Overall, the present framework suggests that Approximate Bayesian Computation may be viewed not only as a methodology for likelihood-free computation, but also as a mathematical theory in which compatibility provides the organizing principle underlying the probabilistic, asymptotic, variational, geometric, and risk-theoretic aspects of simulation-based Bayesian inference.

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